

INFORMATION TECHNOLOGY INSTITUTE

LAB 11

Fully Differential Folded Cascode OTA





Contents

PART 1: gm/ID Design Charts	5
NMOS	5
PMOS.....	5
PART 2: OTA Design	6
Specs & Hand Analysis	6
Sizing	7
Cascode mosfets.....	7
Input pair	9
Sink Nmos CM.....	10
Source PMOS CM.....	11
Sizing Summary table.....	12
PART 3: Open-Loop OTA Simulation (Behavioral CMFB)	13
Schematic of the OTA	13
Difference between input and output of the error amplifier	15
Diff small signal ccs	16
Plot diff gain	16
PART 4: Open-Loop OTA Simulation (Actual CMFB)	18
CMFB circuit design	18
Sensing circuit design.....	18
Error amplifier circuit design.....	19
OTA schematic annotated after real CMFB.....	22
OTA DC OP	23
Biasing of the cascode mosfets.....	24
OTA Common mode level	26



Diff small signal ccs	26
Diff Gain.....	26
Circuit parameters	27
PART 5: Closed Loop Simulation (AC and STB Analysis).....	28
Schematic annotated	28
OTA.....	28
CMFB	29
TB schematic.....	29
Differential closed-loop response	30
VODIFF vs frequency.....	30
In dB	30
Phase	31
Circuit parameters	31
Differential and CMFB loops stability (STB analysis).....	32
Overlaid LG	32
Overlaid Phase.....	32
DIFF circuit parameters.....	33
CM circuit parameters	33
GBW and PM comparison	33
Comment.....	33
LG and GBW comparison between OL and CL.....	33
Comment.....	34
PART 6: Closed Loop Simulation (Transient Analysis).....	34
Differential and CMFB loops stability (transient analysis) (Differential impulse)	34
voutN ,voutp ,vinp, vinn&vocm	34
Tsettle.....	34

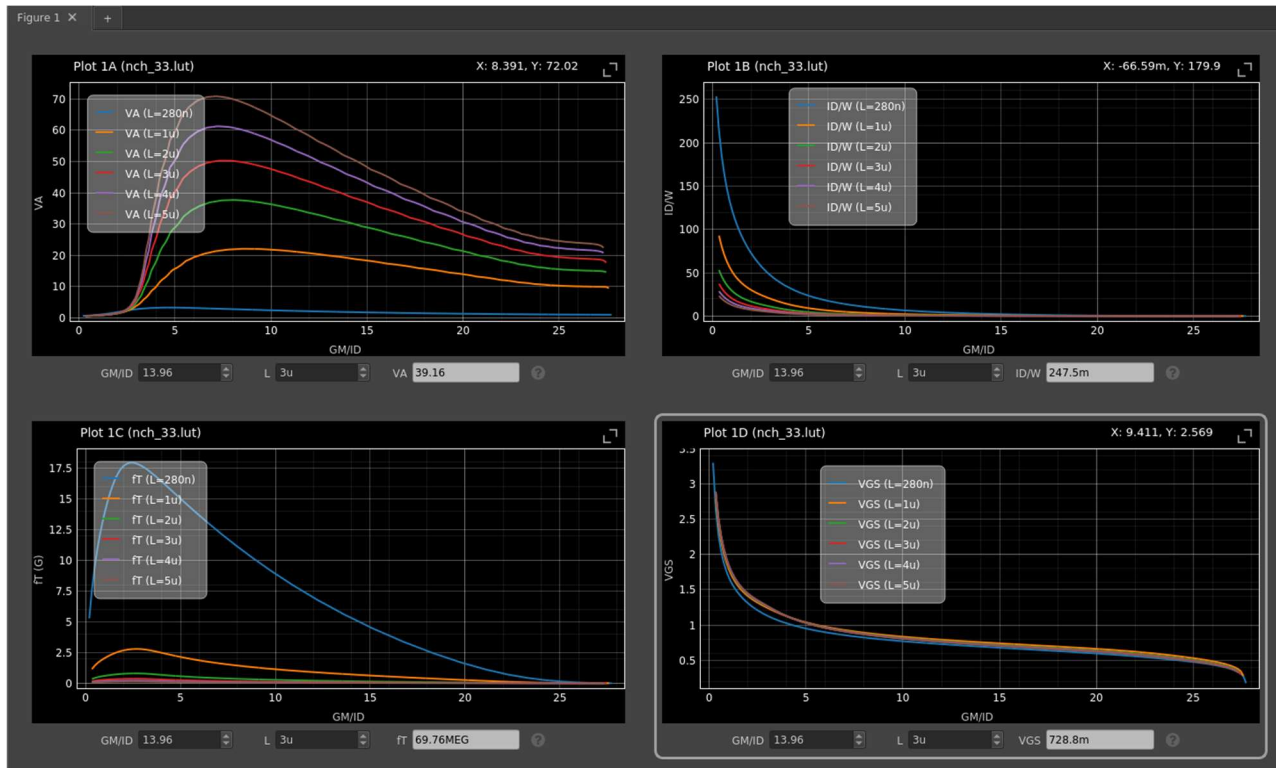


Differential and CMFB loops stability (transient analysis): CM input pulse	35
Output swing	36
VINP, VINN, VOUTP, VOUTN, and VOCM overlaid.....	36
VIDIFF and VODIFF	36

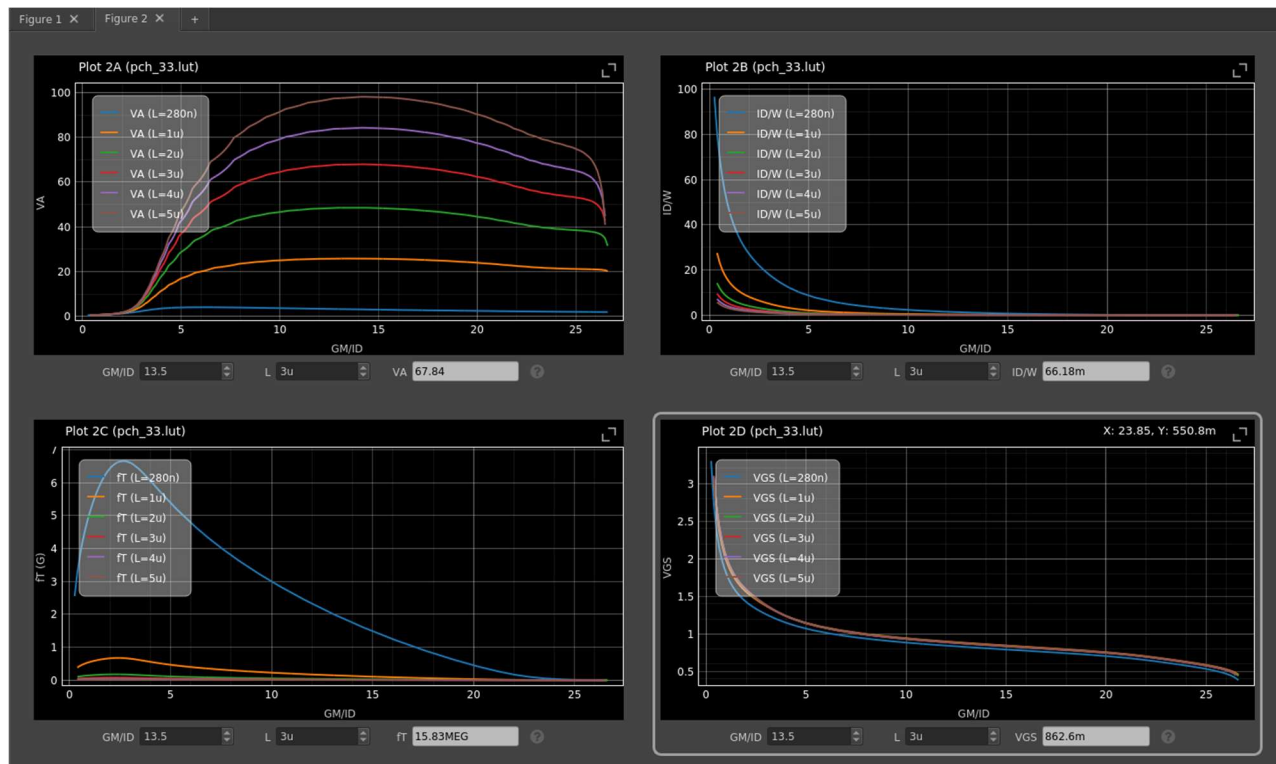


PART 1: gm/ID Design Charts

NMOS



PMOS





PART 2: OTA Design

Specs & Hand Analysis

Technology	0.13um	GF180MCU
Supply voltage	1.2V	2.5V
Closed loop gain	2	2
Phase margin at the required ACL	$\geq 70^\circ$	$\geq 70^\circ$
CM input range – low	≤ 0	≤ 0
CM input range – high	$\geq 0.6V$	$\geq 1V$
Differential output swing	0.6Vpk-to-pk	1.2Vpk-to-pk
Load	500fF	500fF
DC Loop gain	50dB	60dB
CL settling time for 1% error	100ns	100ns

From the Settling time spec we can assume that it is a first order system and calculate the tau

$$BW_{CL}A_{CL} = A_{loaded}BW, A_{CL} = 2$$

$$2BW_{CL} = \frac{C_{in}}{C_f + C_{in}} A_{OL}BW$$

From the settling-time specification we can assume that it is a first order system and calculate the time constant which equals $= \frac{1}{2\pi BW_{CL}}$

$$\tau \approx \frac{t_{s,1\%}}{4.7} \quad \tau \approx \frac{100 \text{ ns}}{5} = 20 \text{ ns}$$

$$BW_{CL} = \frac{1}{2\pi(20 \text{ ns})} = 7.9577 \times 10^6 \text{ Hz}$$

$$2(7.9577 \times 10^6) = \frac{2}{3} \text{ GBW}$$

$$\text{GBW} = 23.873 \text{ MHz} \quad (\text{unloaded})$$



$$GBW = \frac{g_m}{2\pi \left(C_L + \frac{C_{in}C_f}{C_{in} + C_f} \right)}$$

$$23.873 \times 10^6 = \frac{g_m}{2\pi \left(500 \text{ fF} + \frac{2}{3} \text{ pF} \right)}$$

$$g_m \approx 175 \mu\text{S}$$

$$LG = \left(\frac{1}{3} \right) A_{OL} \quad A_{OL} = 3000$$

$$R_{out} = \frac{A_{OL}}{g_m} = \frac{3000}{175 \mu\text{S}} = 17.14 \text{ M}\Omega$$

$$R_{out} = \frac{g_m r_o^2}{4}$$

Assuming all MOSFETs have the same r_o and g_m , and that the sink NMOS receives twice the current.

Sizing

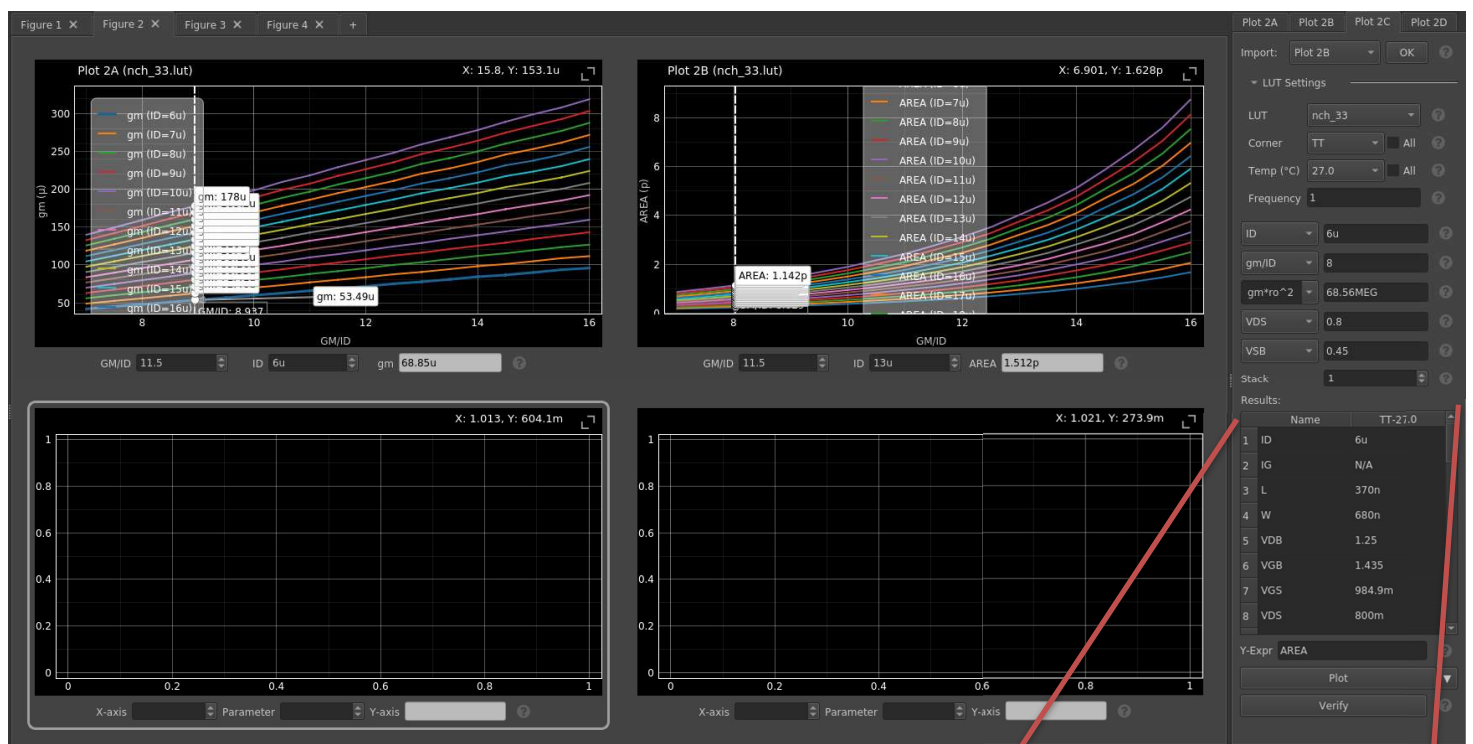
Cascode mosfets

We want the current in the cascode MOSFETs to be as small as possible. Therefore, we set the condition

$$g_m r_o^2 = 68.56 \text{ M}\Omega$$

and sweep g_m/I_D and I_D to find the minimum area and current. We then determine a reasonable R_{out} using the corresponding r_o , while setting the input-transistor r_o equal to the cascode-transistor r_o .

It was found that $I_D = 6 \mu\text{A}$ and $g_m/I_D = 8$ satisfy the condition with a reasonably small area, where r_o is around $1.2 \text{ M}\Omega$



NMOS

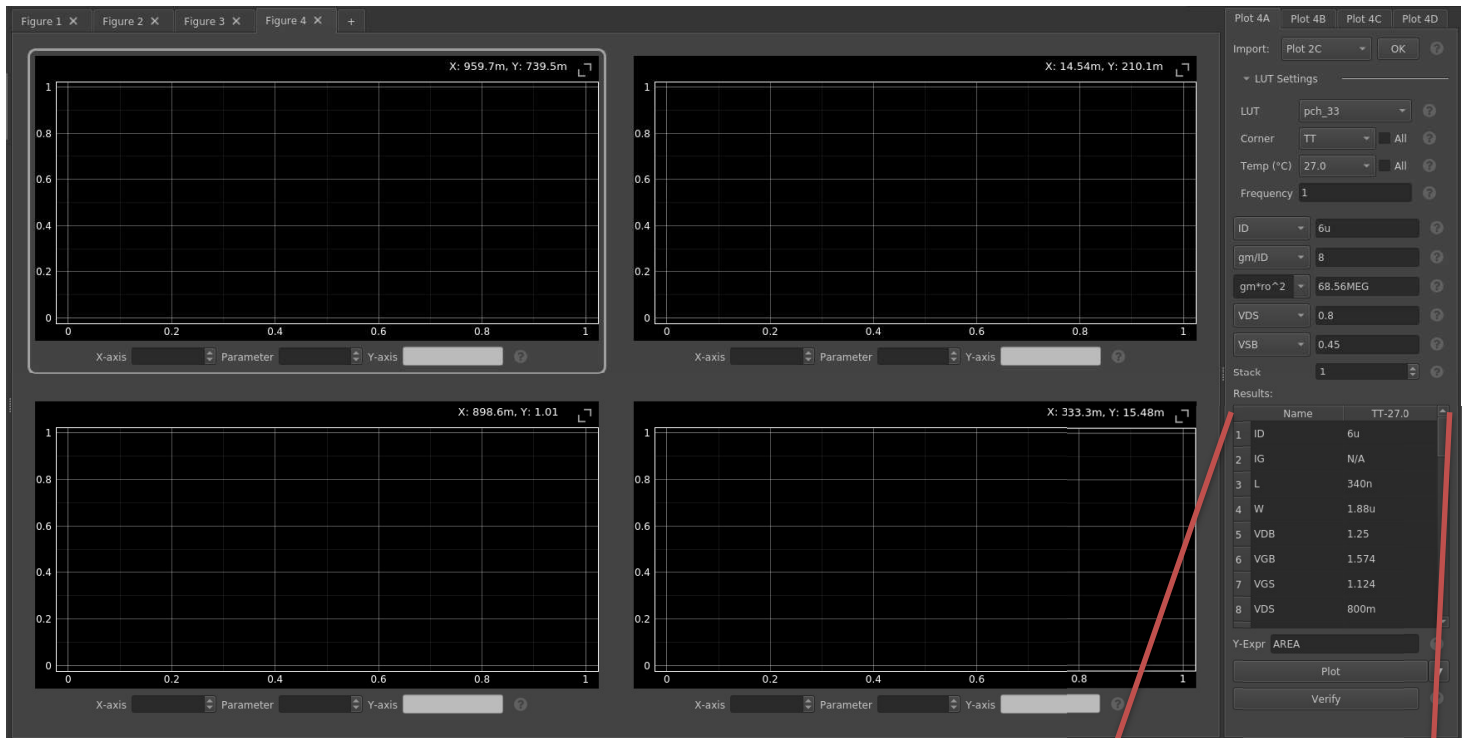
We chose a small current because this branch is responsible for increasing the output resistance:

$$R_{out} = \frac{g_m r_o^2}{4}$$

Since

$$g_m \propto \sqrt{I_D} \text{ and } r_o \propto \frac{1}{I_D},$$

decreasing the current increases R_{out} , and consequently increases the gain, while the GBW is not affected because the input g_m is what affects the GBW.



PMOS

Name		TT-27.0
1	ID	6u
2	IG	N/A
3	L	340n
4	W	1.88u
5	VDB	1.25
6	VGB	1.574
7	VGS	1.124
8	VDS	800m
Y-Expr		AREA

Input pair

We will use an PMOS input pair due the low input range .

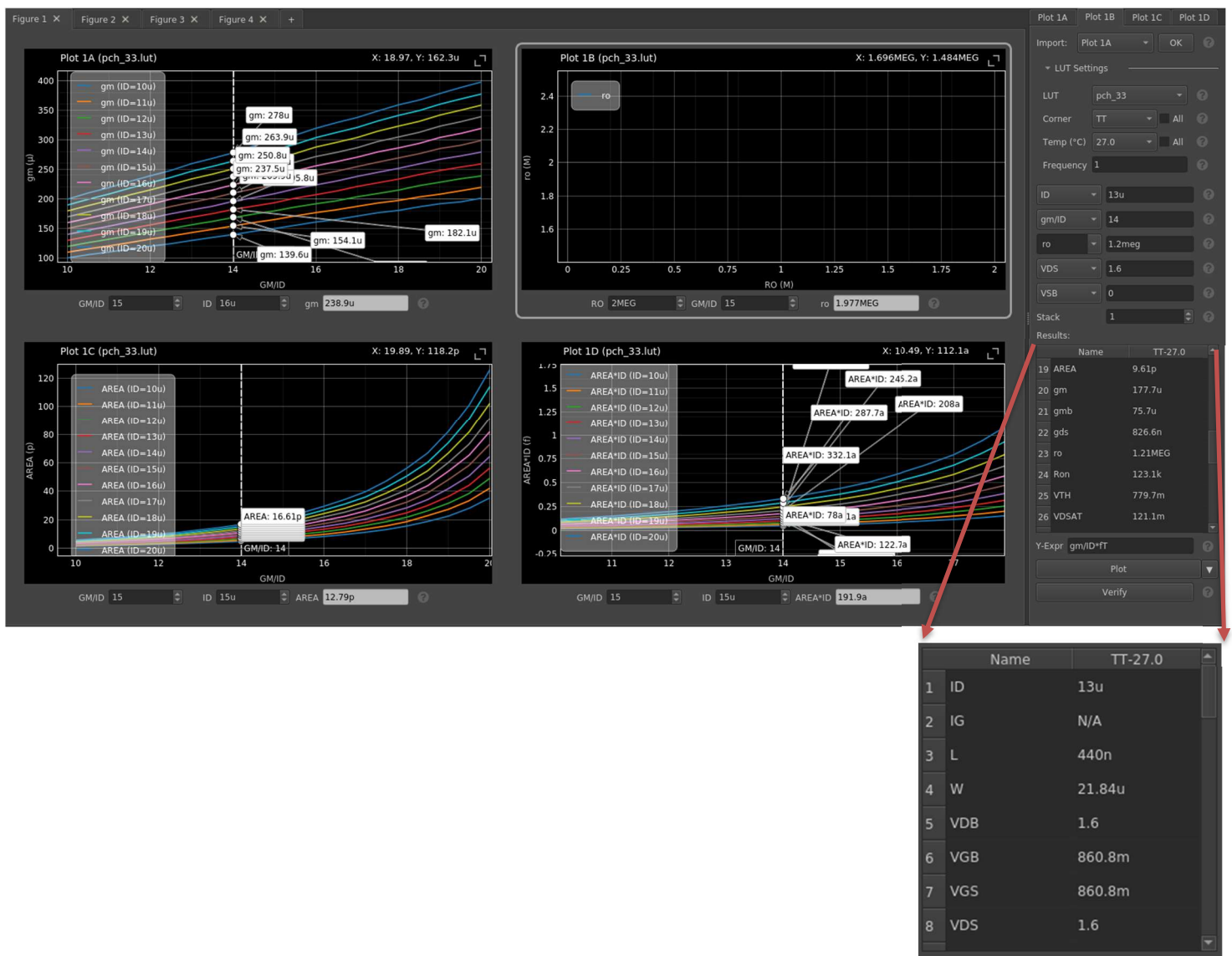
Next, we set the input-pair output resistance to

$$r_o = 12 \text{ M}\Omega$$

and swept g_m/I_D and I_D to achieve

$$g_m = 175 \mu\text{S}$$

with the minimum possible area. The suitable values were found to be g_m/I_D 14 and current 13u



Sink Nmos CM

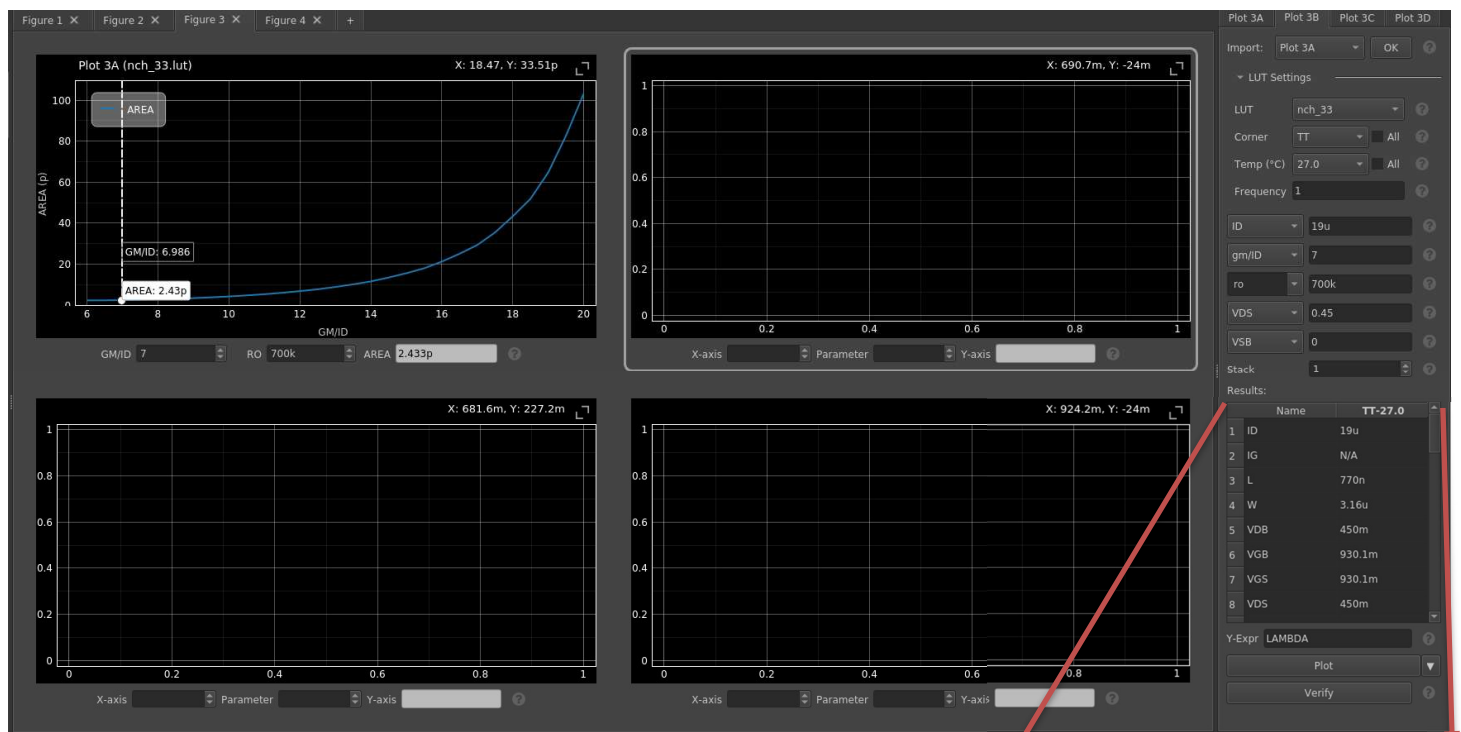
Next, we size the sink NMOS current mirror. We want its output resistance to be at least half that of the remaining devices so that our design assumption remains valid. Therefore, assume

$$r_o = 700 \text{ k}\Omega.$$

Its current is known:

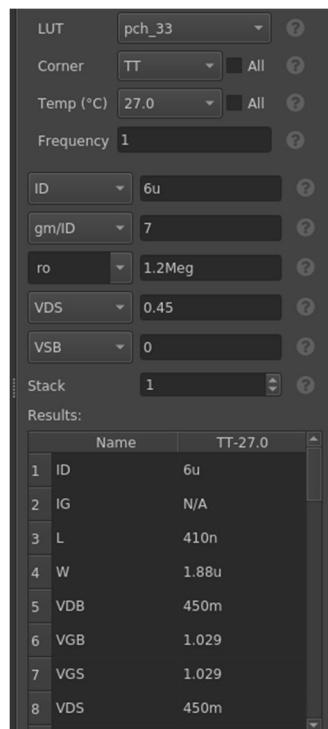
$$I_D = 13 \mu\text{A} + 6 \mu\text{A} = 19 \mu\text{A}, \text{ And will be around } V_{DS} = 450 \text{ mV}.$$

We then minimize its area. The required conditions are achievable at $g_m/I_D = 7$



Source PMOS CM

Same gm/id as the nmos sink but the $r_o = 1.2 \text{ M}\Omega$.



Name		TT-27.0
1	ID	19u
2	IG	N/A
3	L	770n
4	W	3.16u
5	VDB	450m
6	VGB	930.1m
7	VGS	930.1m
8	VDS	450m



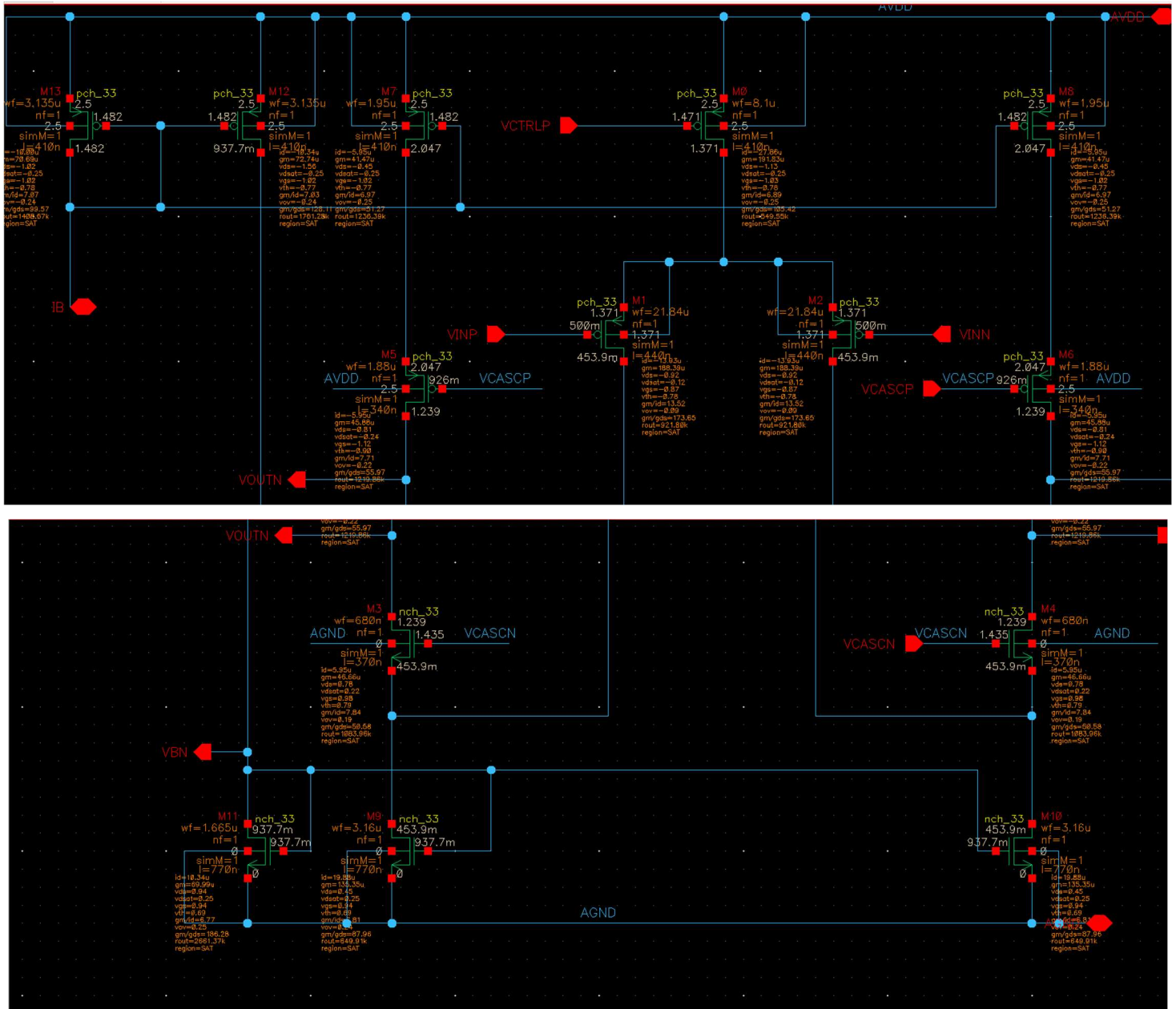
Sizing Summary table

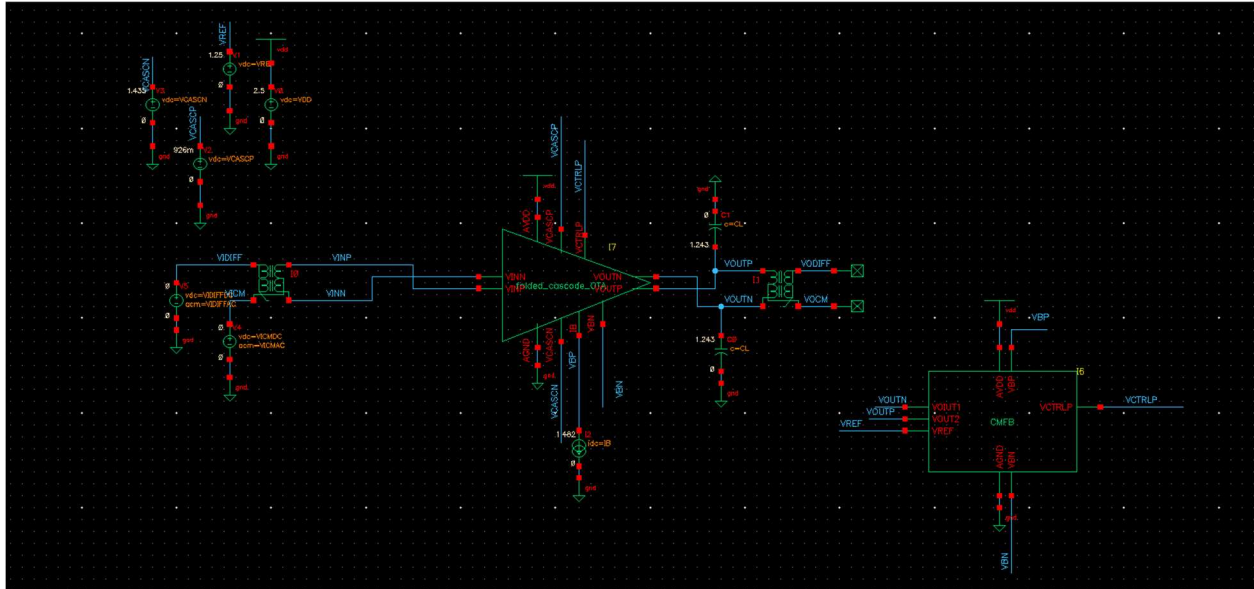
Transistor(s)	Function	Sizing	DC operating point
M₀	CMFB-controlled PMOS source	$W = 8.10 \mu\text{m}$ $L = 0.410 \mu\text{m}$ $g_{mI_D} = 6.89 \text{ V}^{-1}$ $ I_D = 27.86 \mu\text{A}$	$V_{GS} = -1.03 \text{ V}$ $V_{TH} = -0.78 \text{ V}$ $V_{DS} = -1.13 \text{ V}$
M₁, M₂	PMOS input pair	$W = 21.84 \mu\text{m}$ $L = 0.440 \mu\text{m}$ $g_{mI_D} = 13.52 \text{ V}^{-1}$ $ I_D = 13.93 \mu\text{A}$	$V_{GS} = -0.87 \text{ V}$ $V_{TH} = -0.78 \text{ V}$ $V_{DS} = -0.92 \text{ V}$
M₃, M₄	NMOS cascodes	$W = 0.680 \mu\text{m}$ $L = 0.370 \mu\text{m}$ $g_{mI_D} = 7.84 \text{ V}^{-1}$ $I_D = 5.95 \mu\text{A}$	$V_{GS} = 0.98 \text{ V}$ $V_{TH} = 0.79 \text{ V}$ $V_{DS} = 0.78 \text{ V}$
M₅, M₆	PMOS cascodes	$W = 1.88 \mu\text{m}$ $L = 0.340 \mu\text{m}$ $g_{mI_D} = 7.71 \text{ V}^{-1}$ $ I_D = 5.55 \mu\text{A}$	$V_{GS} = -1.12 \text{ V}$ $V_{TH} = -0.90 \text{ V}$ $V_{DS} = -0.81 \text{ V}$
M₇, M₈	Source PMOS current-mirror outputs	$W = 1.95 \mu\text{m}$ $L = 0.410 \mu\text{m}$ $g_{mI_D} = 6.97 \text{ V}^{-1}$ $ I_D = 5.95 \mu\text{A}$	$V_{GS} = -1.02 \text{ V}$ $V_{TH} = -0.77 \text{ V}$ $V_{DS} = -0.45 \text{ V}$
M₉, M₁₀	Sink NMOS current-mirror outputs	$W = 3.16 \mu\text{m}$ $L = 0.770 \mu\text{m}$ $g_{mI_D} = 6.81 \text{ V}^{-1}$ $I_D = 19.88 \mu\text{A}$	$V_{GS} = 0.94 \text{ V}$ $V_{TH} = 0.69 \text{ V}$ $V_{DS} = 0.45 \text{ V}$
M₁₁	Sink NMOS mirror reference	$W = 1.665 \mu\text{m}$ $L = 0.770 \mu\text{m}$ $g_{mI_D} = 6.77 \text{ V}^{-1}$ $I_D = 10.34 \mu\text{A}$	$V_{GS} = 0.94 \text{ V}$ $V_{TH} = 0.69 \text{ V}$ $V_{DS} = 0.94 \text{ V}$
M₁₂	Source PMOS mirror output/bias branch	$W = 3.135 \mu\text{m}$ $L = 0.410 \mu\text{m}$ $g_{mI_D} = 7.03 \text{ V}^{-1}$ $ I_D = 10.34 \mu\text{A}$	$V_{GS} = -1.02 \text{ V}$ $V_{TH} = -0.77 \text{ V}$ $V_{DS} = -1.56 \text{ V}$
M₁₃	Source PMOS mirror reference	$W = 3.135 \mu\text{m}$ $L = 0.410 \mu\text{m}$ $g_{mI_D} = 7.07 \text{ V}^{-1}$ $ I_D = 10.00 \mu\text{A}$	$V_{GS} = -1.02 \text{ V}$ $V_{TH} = -0.78 \text{ V}$ $V_{DS} = -1.02 \text{ V}$



PART 3: Open-Loop OTA Simulation (Behavioral CMFB)

Schematic of the OTA





As

Maximum $v_{out} = 2.5 - (450\text{mV} + 250) = 1.8$, the V_{ds} of the Pmos source is set by the v_{gs} of the cascode $= 450\text{mV}$ and the v_{dsat} of the cascode $= 250\text{mV}$

$$\frac{1.8+0.67}{2} = 1.235\text{v}$$

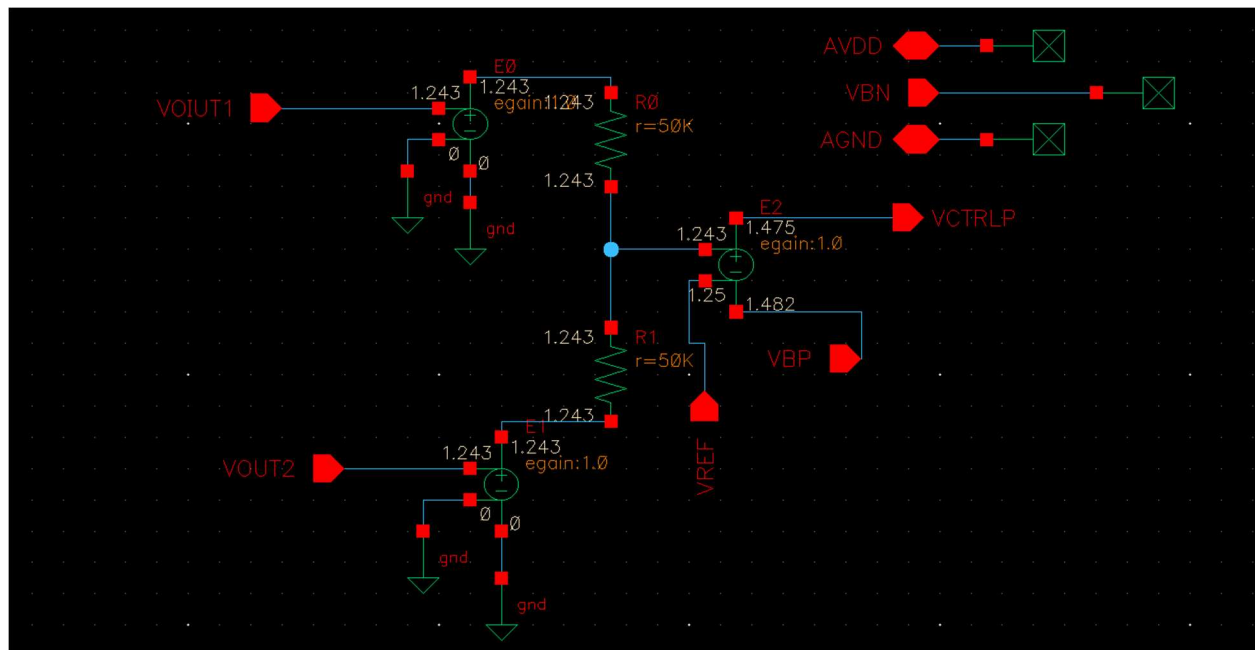
So V_{ref} is approximately 1.25

The CM at the output of the OTA is 1.239v

When the OTA amplifier screen shot was taken it was set to 1.235 v but then I setted Vref to 1.25v this why the output level in the test bench is a little different at 1.243v .



Difference between input and output of the error amplifier



The differential input is $1.243 - 1.25 = -0.007\text{V}$

The relation is $V_{out} - V_{outq} = v_{in+} - v_{in-}$ the gain here is unity and $V_{outq} = V_{BP} = 1.482$ that's why we did not need a large gain as V_{outq} is already near the v_{out} needed value



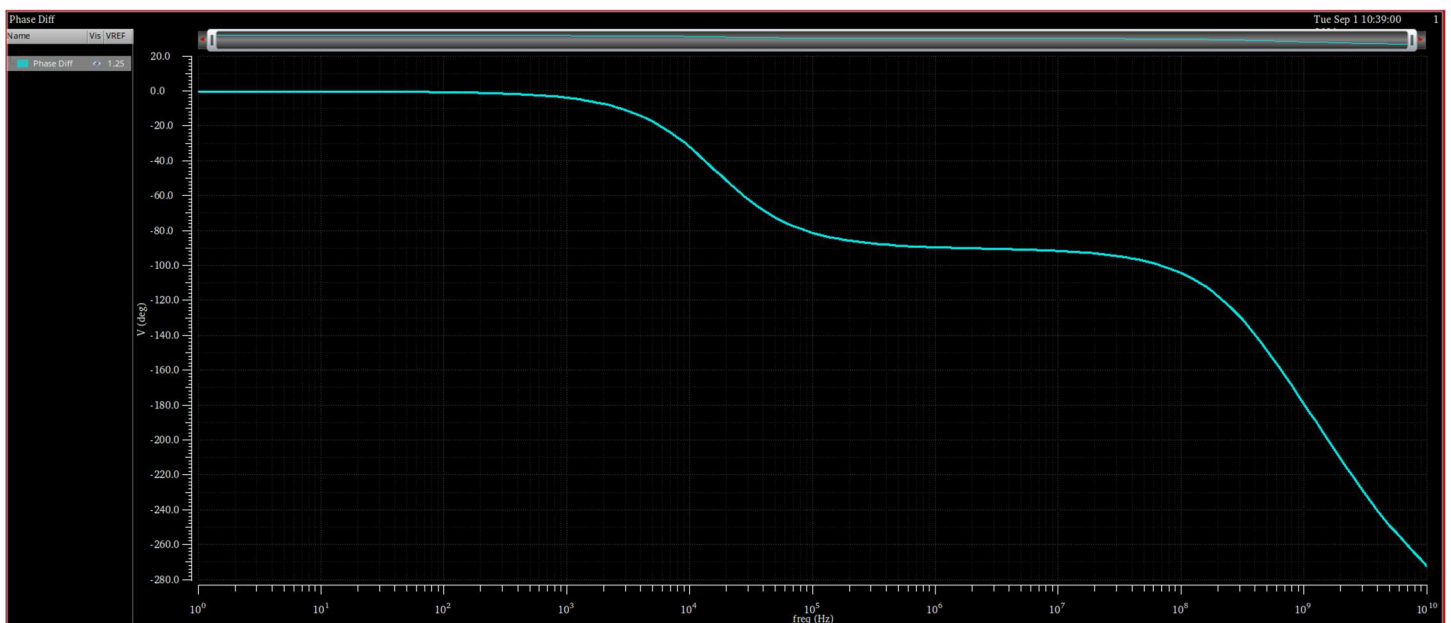
Diff small signal ccs

Plot diff gain

Magnitude in dB vs frequency



Phase vs frequency





Circuit parameters

lab11:TB:1	Ao Diff	3.57k
lab11:TB:1	Ao Diff dB	71.05
lab11:TB:1	BW Diff	15.97k
lab11:TB:1	UGF Diff	56.84M
lab11:TB:1	GBW Diff	57M
lab11:TB:1	PM Diff	82.02

Hand Analysis

$$g_{m,in} = 188.39 \mu S, g_{m,cas} = 45.88, r_o = 1.21988 M$$

$$R_{out} \approx \frac{g_{m,c} r_{o,c}^2}{4}$$

$$R_{out} \approx \frac{(45.88 \mu S)(1.21988 M\Omega)^2}{4} = 17.07 M\Omega$$

$$A_{0,diff} \approx g_{m,in} R_{out}$$

$$A_{0,diff,dB} = 20 \log_{10}(3.216 \times 10^3) = 70.15 dB$$

$$BW_{diff} \approx \frac{1}{2\pi R_{out} C_L}$$

$$BW_{diff} \approx \frac{1}{2\pi(17.07 M\Omega)(500 fF)} = 18.65 kHz$$

$$GBW_{diff} \approx A_{0,diff} BW_{diff}$$

$$GBW_{diff} \approx (3.216 \times 10^3)(18.65 kHz) = 59.97 MHz$$

$$f_{UGF} = BW_{diff} \sqrt{A_{0,diff}^2 - 1} \approx GBW_{diff}$$

$$PM \approx 180^\circ - \tan^{-1} \left(\frac{f_{UGF}}{BW_{diff}} \right)$$

$$PM \approx 180^\circ - \tan^{-1} \left(\frac{59.97 MHz}{18.65 kHz} \right) \approx 90^\circ$$

Comparison

Parameter	Hand calculation	Simulation
$A_{0,diff}$	3.216 kV/V	3.57 kV/V
$A_{0,diff,dB}$	70.15 dB	71.05 dB
BW_{diff}	18.65 kHz	15.97 kHz
f_{UGF}	59.97 MHz	56.84 MHz
GBW_{diff}	59.97 MHz	57.00 MHz
Phase margin	90.02°	82.02°



PART 4: Open-Loop OTA Simulation (Actual CMFB)

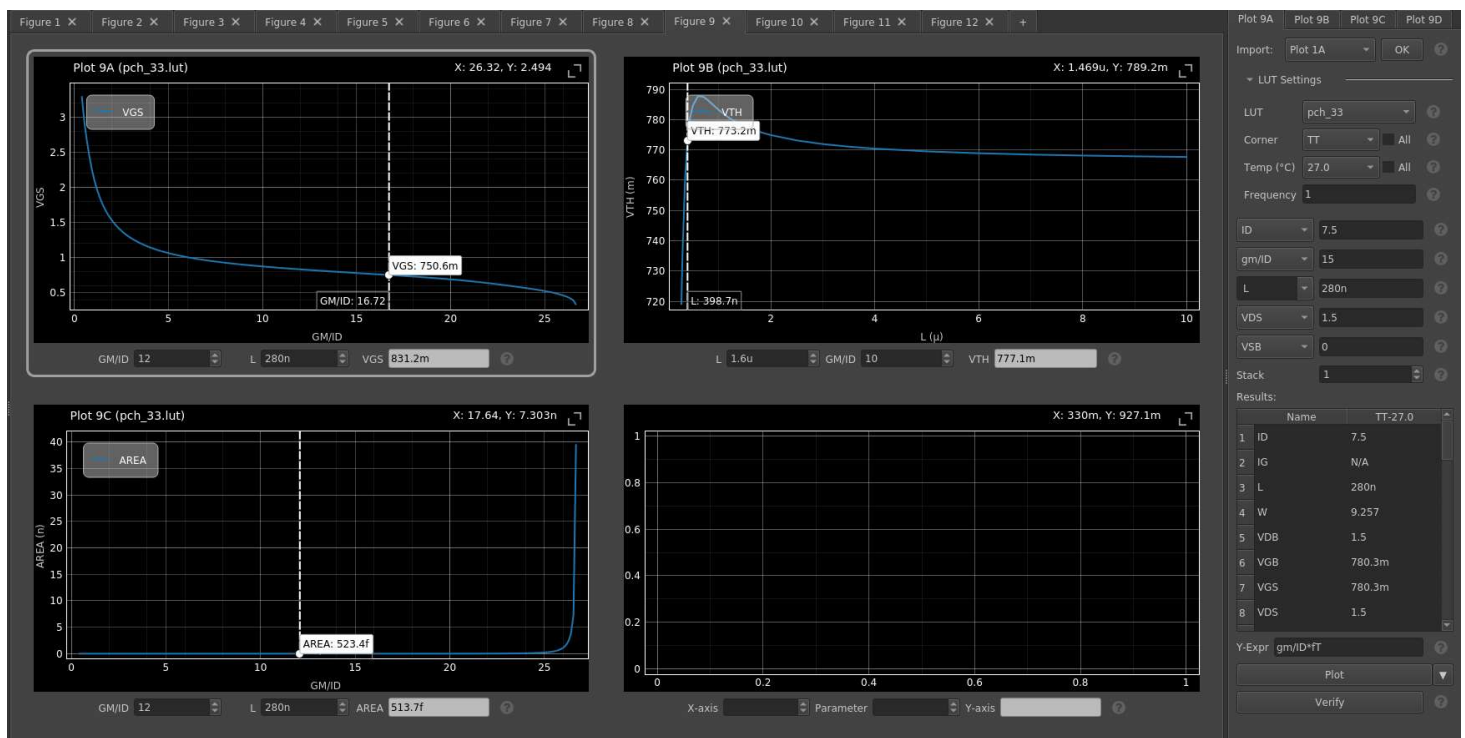
CMFB circuit design

Sensing circuit design

Assume a given resistors value of 50k and V_{diff} peak to peak is 1.2 so the maximum current that could steer in the sensing circuit is around $\frac{0.6}{100k} = 6 \mu A$ so the current should be more than that in each branch to avoid starvation of the CD MOS I choose $7.5 \mu A$

CD MOSfets

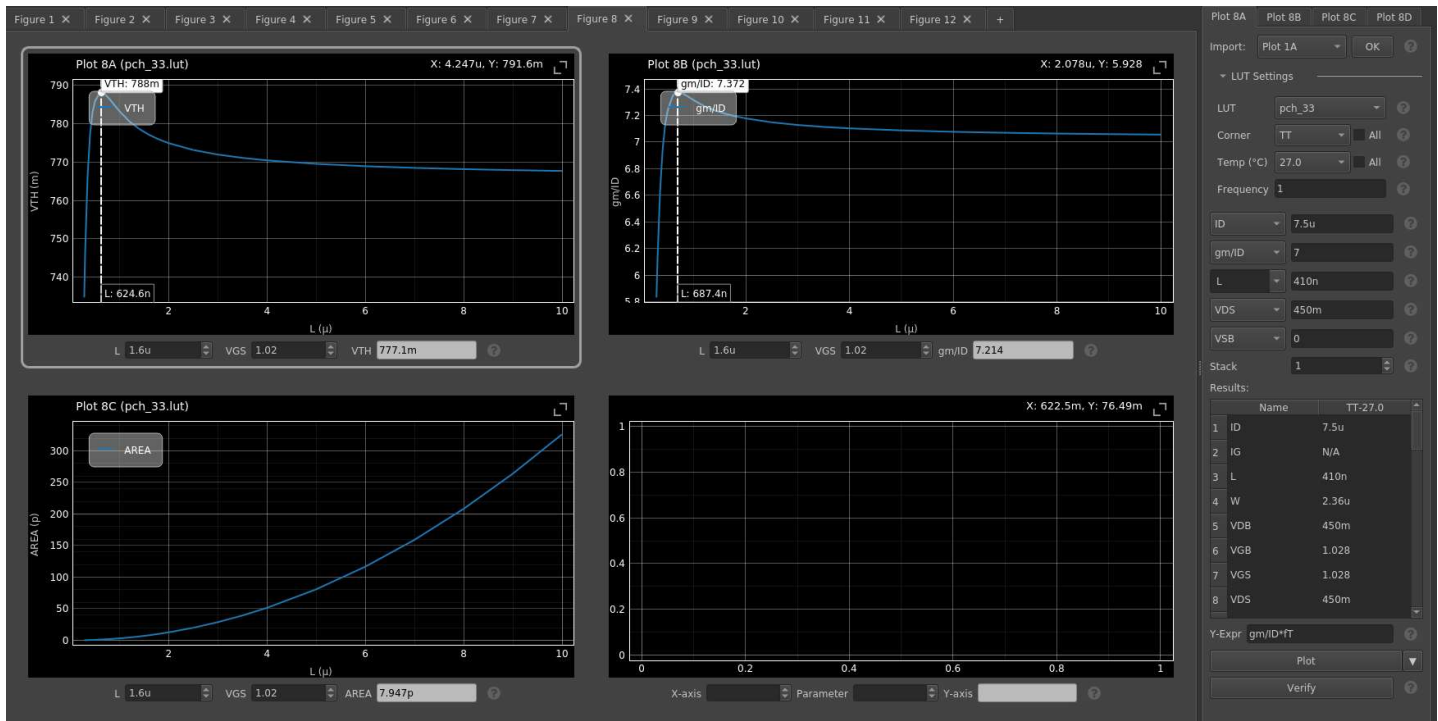
For them I choose a relatively large gm/id at 15 to decrease the V_{gs} as much as possible and there r_o are not important so I choose minimum L so that the V_{th} is at its minimum thus there V_{GS}





Sources MOSfets

Should have the same g_m/i_d as other PMOS sources and same L as they are biased with same V_{GS} as the rest current mirror devices



Due to this relatively large g_m/i_d the upper maximum swing is a bit compromised but still achieves the required spec and we will also change the V_{OCM} to the middle of the maximum swing .

Error amplifier circuit design

We want a small gain thus we will use diode connected load devices we also want no additional H_{IN} for the stability of the CMFB as we don't want to compensation caps . starting with current we want a reasonable current so we will choose 4 μ A for each branch .



Input devices

For them we want a g_m larger than the load so they will be biased at higher g_m but to keep the area minimum as we want benefit from the gain that much we will use a relatively reasonable g_m/id at 12 & small L at 300n to minimize the area

Name	TT-27.0
1 ID	4u
2 IG	N/A
3 L	300n
4 W	1u
5 VDB	1.025
6 VGB	1.289
7 VGS	838.8m
8 VDS	575m

Y-Expr: $g_m/ID \cdot TT$

Buttons: Plot, Verify

PMOS diode connected load

The VGS will already be set by 2.5-VBP to keep V_{outq} around the biasing voltage of the controlled PMOS source

Name	TT-27.0
1 ID	4.041u
2 IG	N/A
3 L	300n
4 W	640n
5 VDB	1.025
6 VGB	1.025
7 VGS	1.025
8 VDS	1.025

Y-Expr: $g_m/ID \cdot TT$

Buttons: Plot, Verify



SINK NMOS

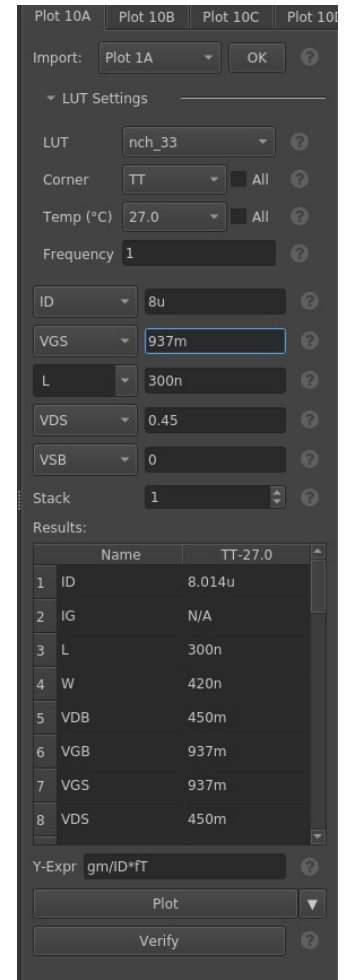
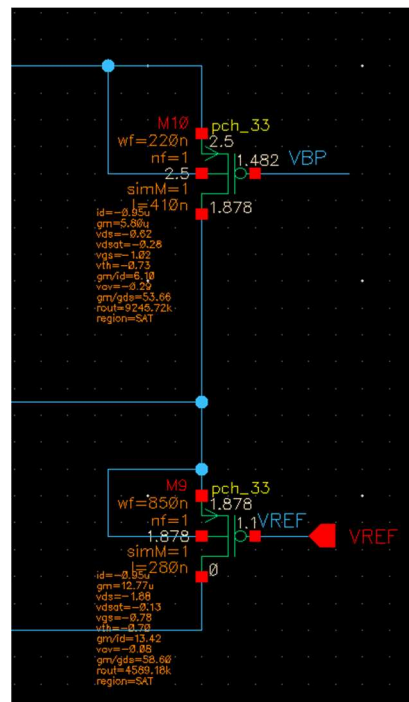
I sized using value of the VBN as the VGS to minimize it could have kept the same L as the other sinking NMOS but to minimize the area I choose to have a different as the other sinking nmos have a relatively large L

So I choose the L around the 300n and the current and the VGS is given

VREF shifting branch

I choose to bias with a 1 uA and the VGS of the input pmos should be the same the CD for the Common input so relatively same L = 300n and the PMOS source same L as the other source and the W is set for a current of 1uA

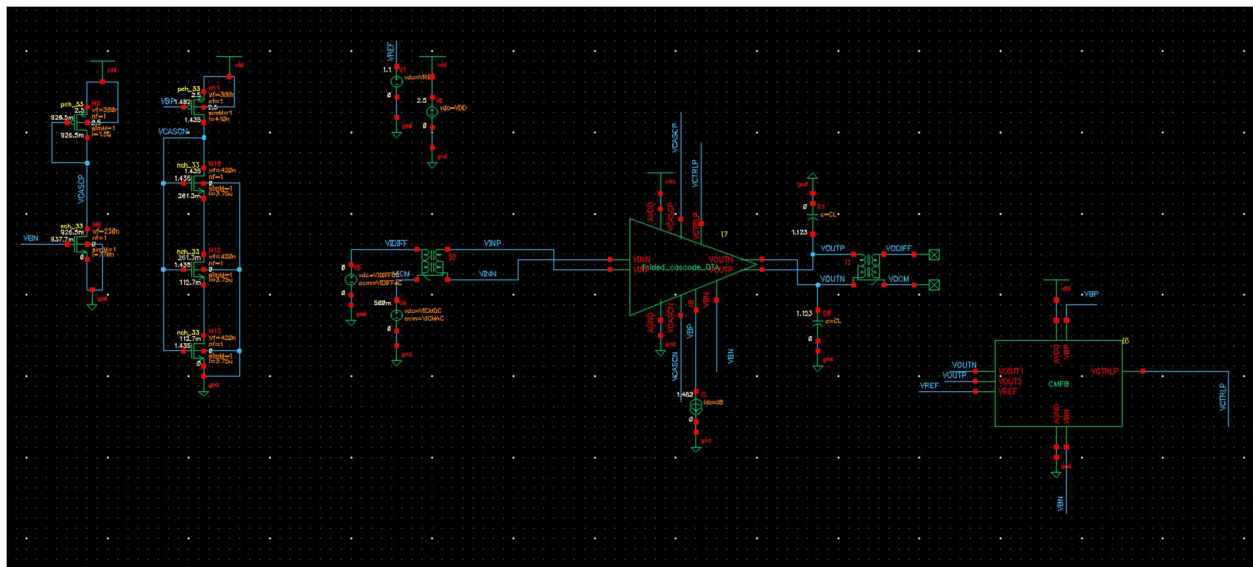
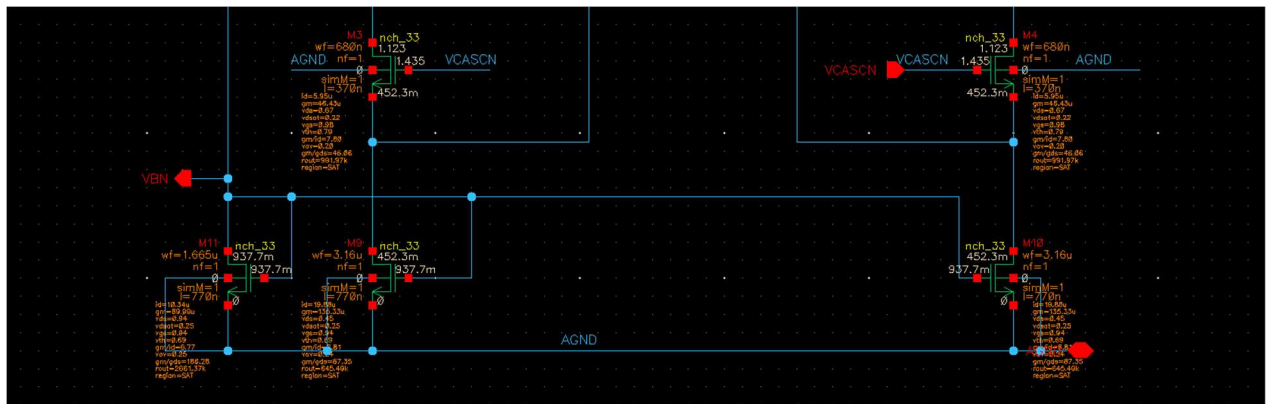
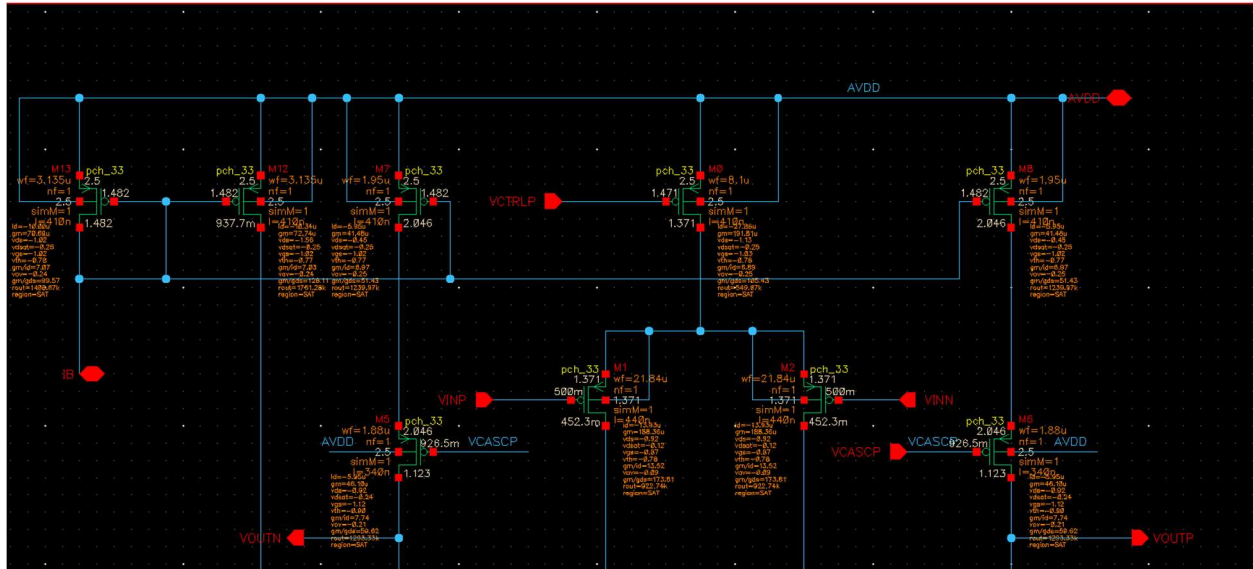
to decrease the current a bit the the lower PMOS L was set at 280n which did not affect the VGS



Schematic annotated



OTA DC OP

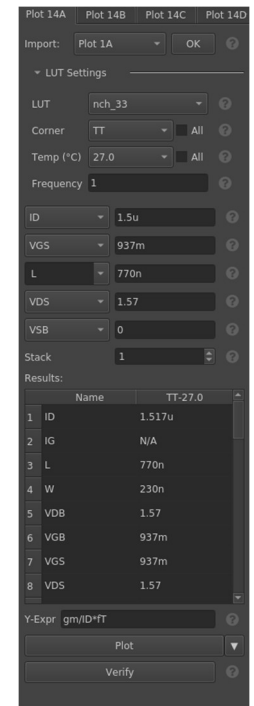
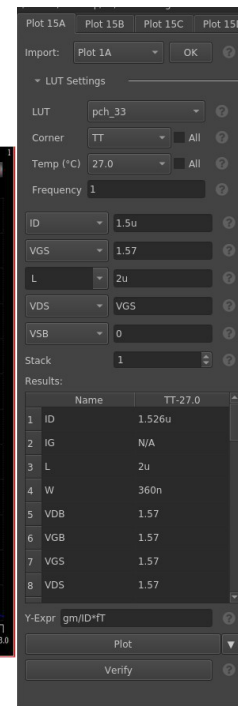
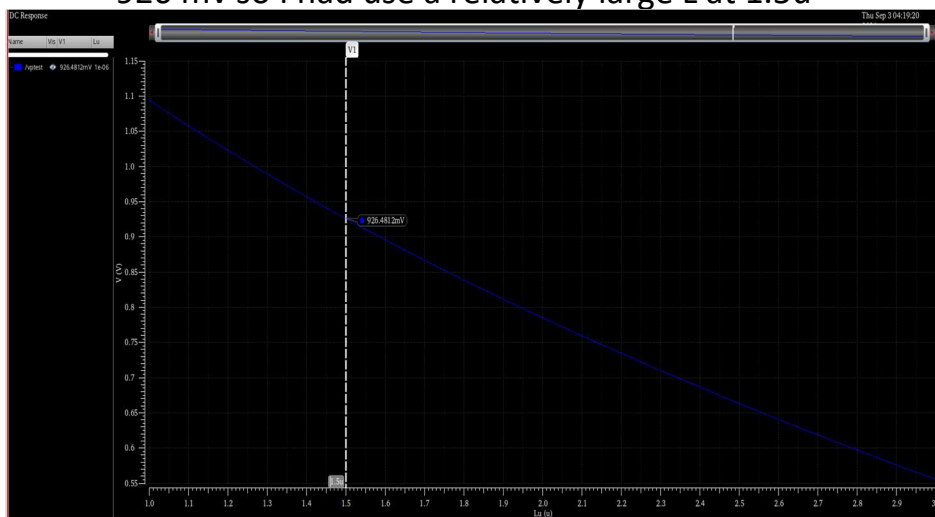
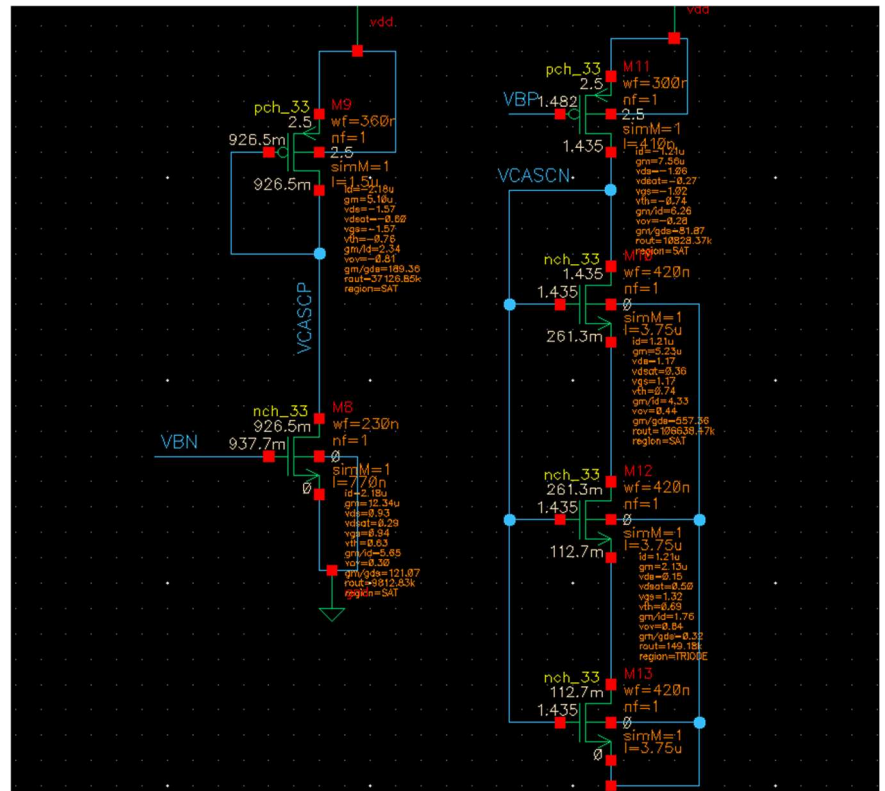




Biasing of the cascode mosfets

I used this biasing circuit to generate the needed voltage at the gate of each cascode mosfet

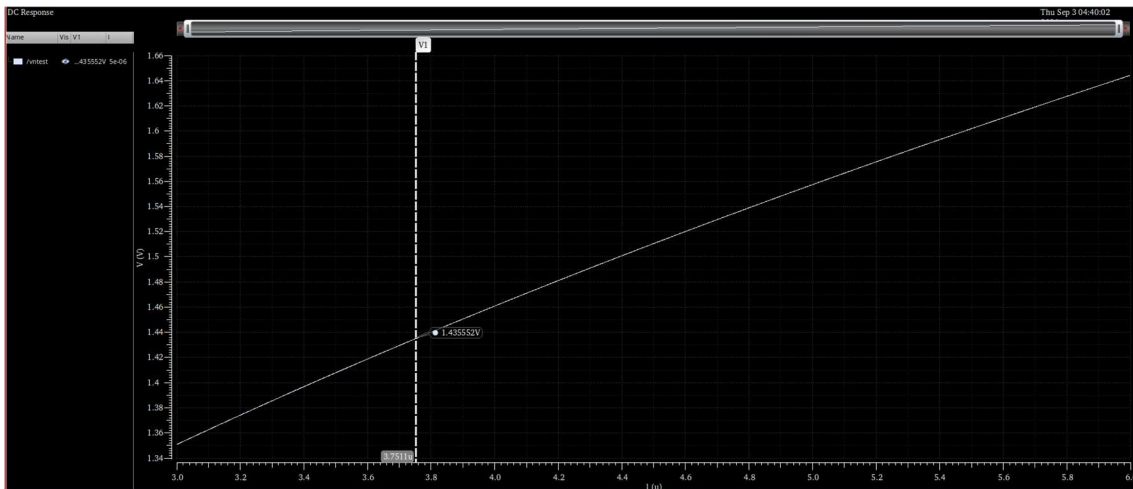
PMOS:current used in pmos cascode biasing circuit is 1.5u which turned to 2u in simulation so I had to decrease the current in the other to stay within the 5u current budget and the pmos cascode had the higher current budget due to the nmos sink which I couldn't decrease its width more than this as its L is the same as the other sinks and the W is scaled and I used a diode connected PMOS to generate around 926 mv so I had use a relatively large L at 1.5u



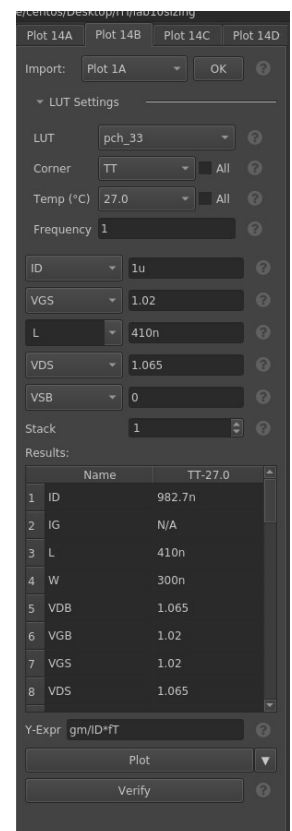
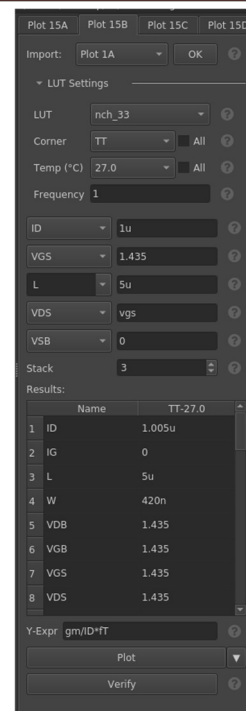


I had to fine tune the L of the pmos through sweeping on it generate exactly the voltage I need so initially it was at 2u then by sweeping the L the correct value was at 1.5u

NMOS: current used was 1u turned in the simulation to 1.16u the as when designing the PMOS cascode biasing circuit the difference is that to achieve the needed voltage which is 1.435v at valid width I needed to use a stack of three as without using it the L will be more than the maximum L and then I swept the L to fine tune



The voltages that we need are the desirable VDS for the sink or source mosfets in the cascode branch add to it the VGS need for the cascode





OTA Common mode level

It is around 1.123v as the Vref is set 1.1 and due to the CMFB which is negative the difference between the CM and Vref is minimal

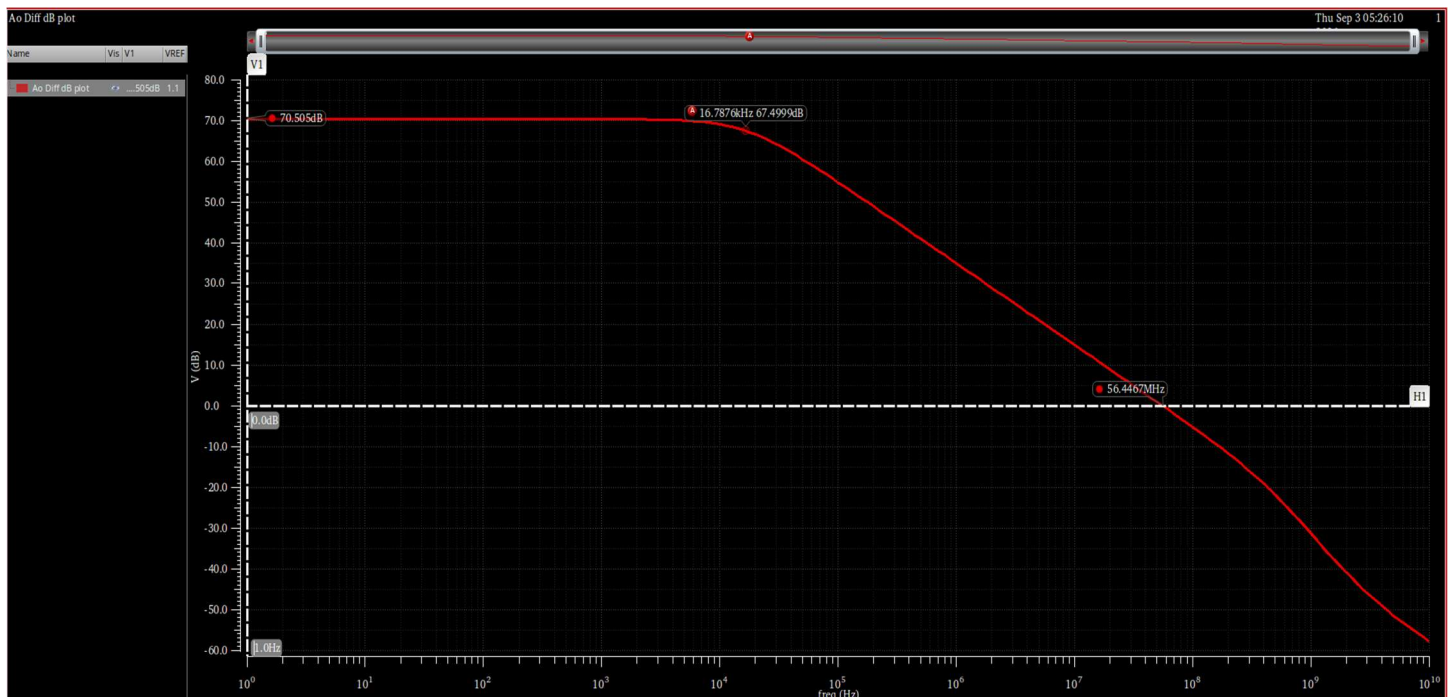
Error amplifier gain is $g_{mn}/g_{mp} * 0.5 = 50/24 * 0.5 = 1$

The relation is $V_{out} - V_{outq} = v_{in+} - v_{in-}$ which is the same as previous behavioral circuit the difference is one the gain as the V_{outq} is also = VBP and in my case the gain is also =1 so it is the same

Diff small signal ccs

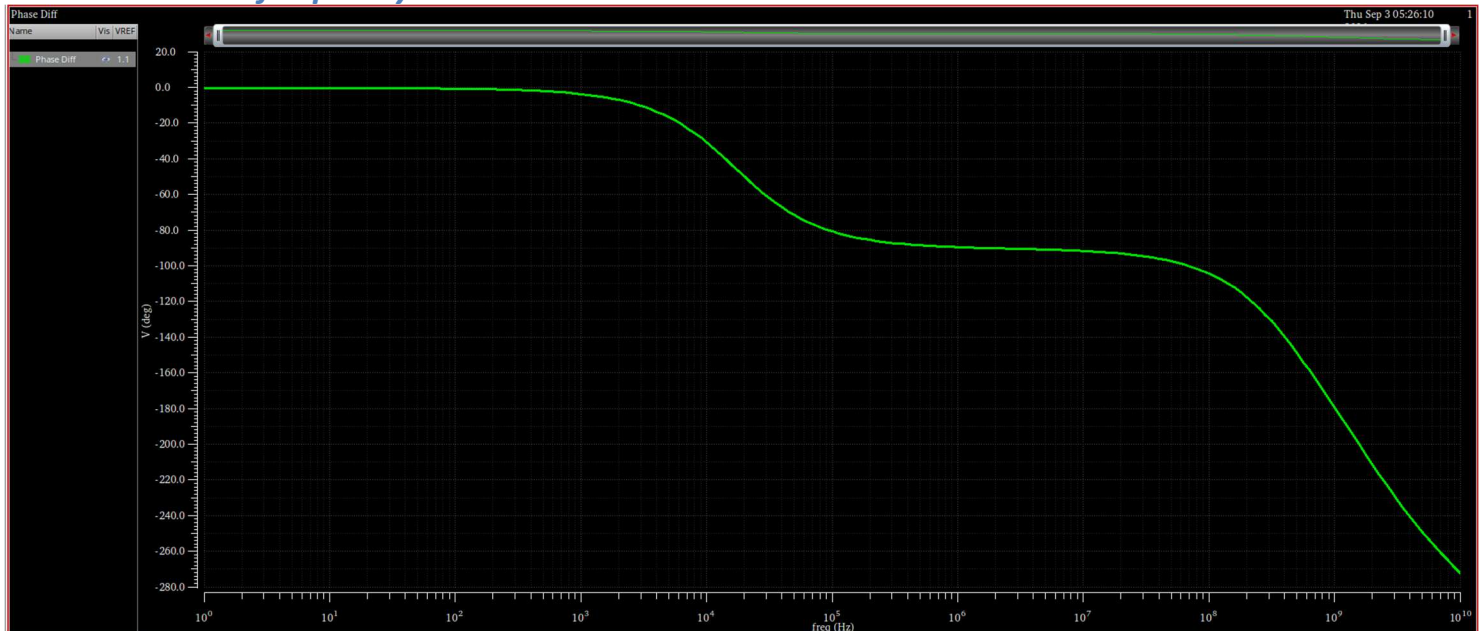
Diff Gain

Magnitude in dB vs frequency





Phase vs frequency



Circuit parameters

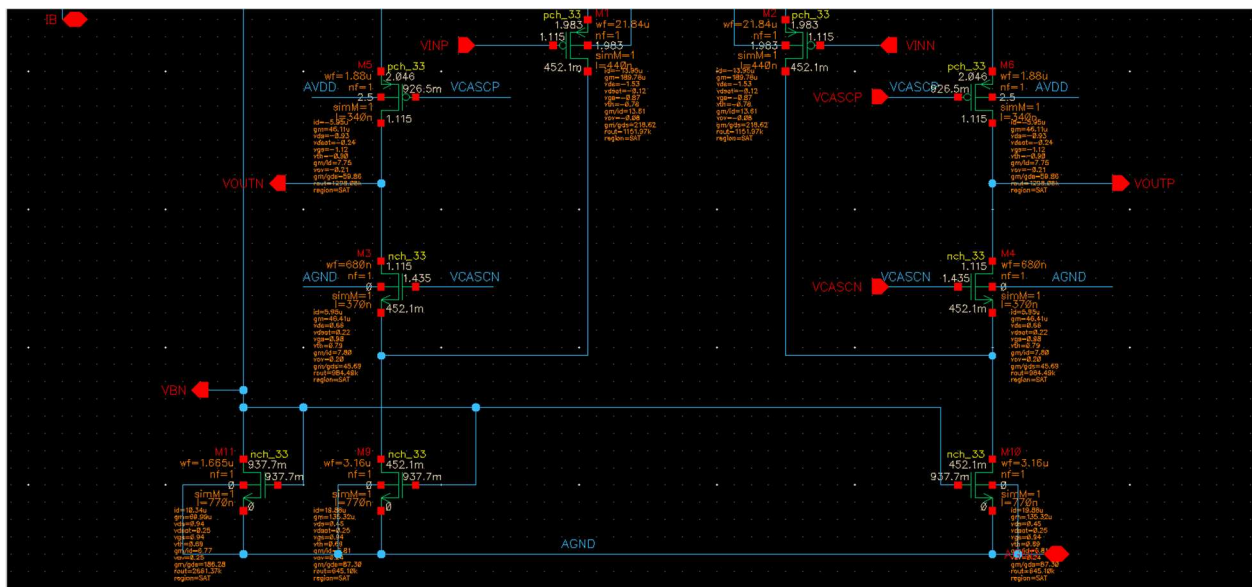
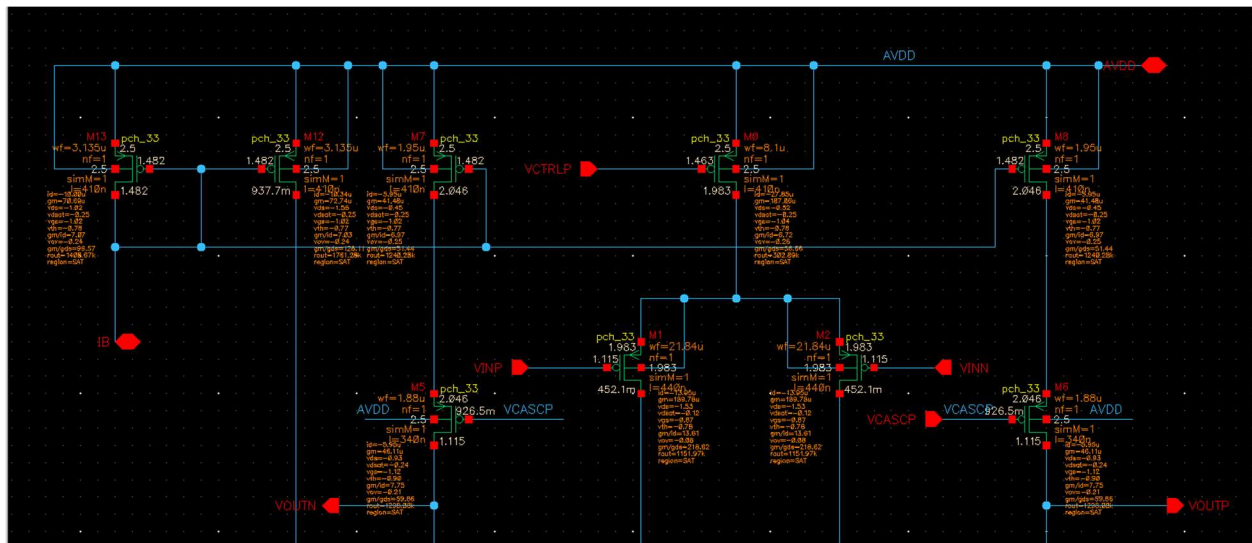
lab11:TB:1	Ao Diff	3.352k
lab11:TB:1	Ao Diff dB	70.51
lab11:TB:1	BW Diff	16.91k
lab11:TB:1	UGF Diff	56.46M
lab11:TB:1	GBW Diff	56.69M
lab11:TB:1	PM Diff	82.04

The hand analysis and the circuit parameters won't be different from the CMFB behavioral circuit was used as the only change is a slight change in the VDS of the NMOS cascode which won't affect the ROUT that much

PART 5: Closed Loop Simulation (AC and STB Analysis)

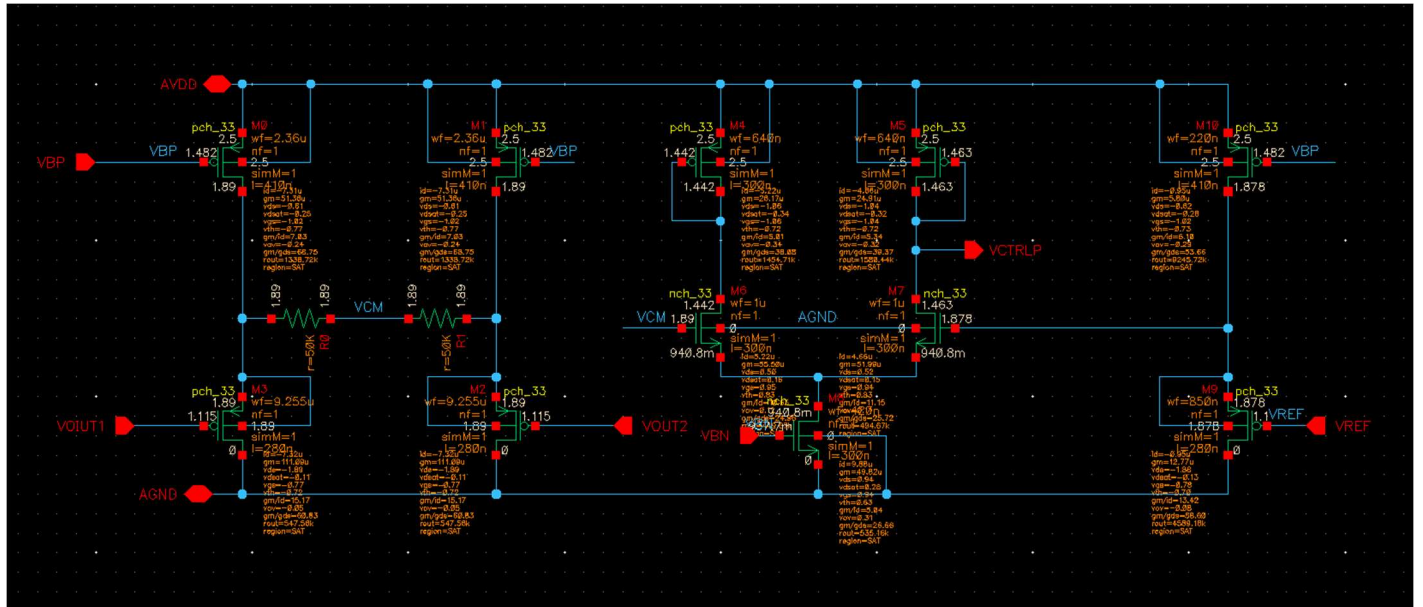
Schematic annotated

OTA

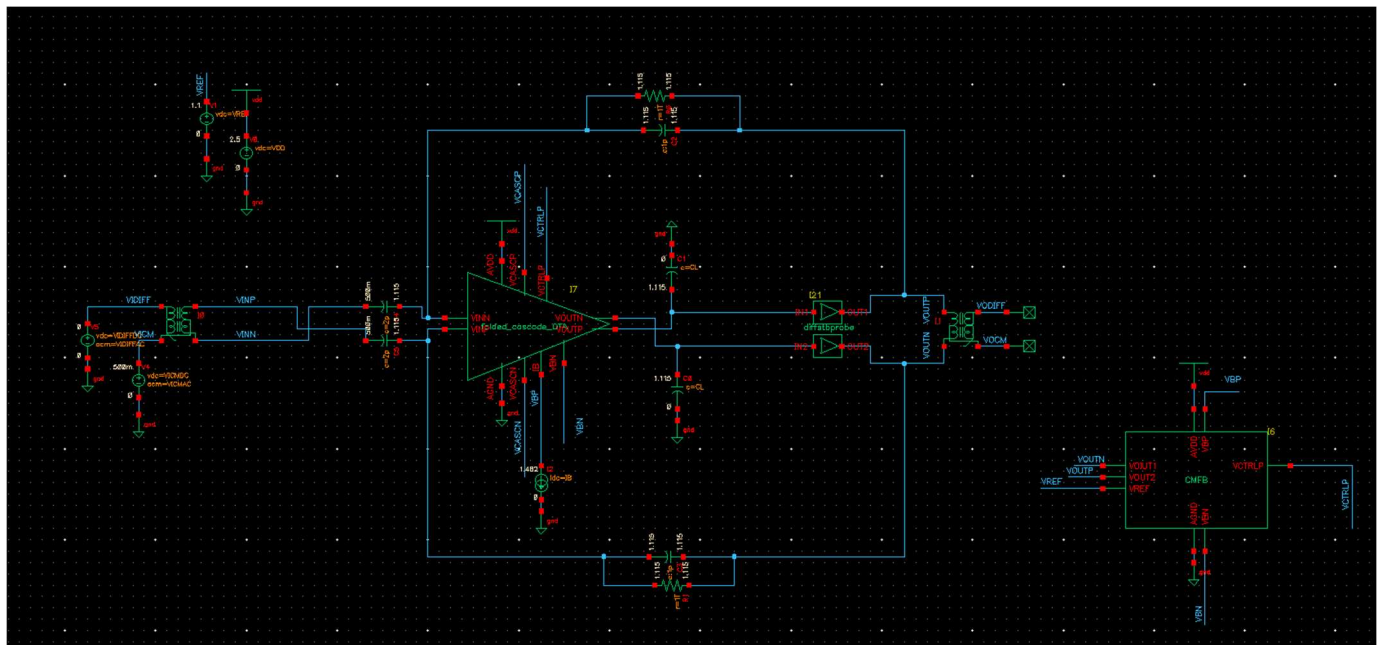




CMFB



TB schematic



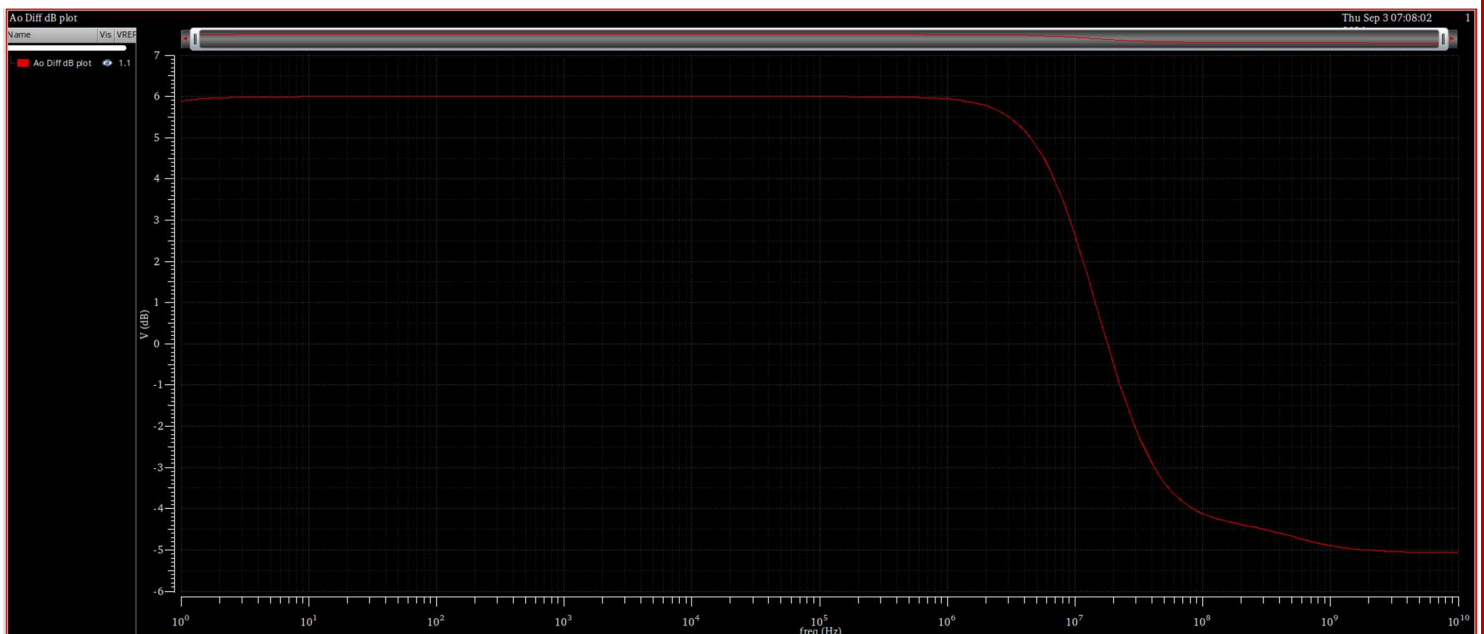


The CM level at the input is the same as input = 1.1v as the input caps block external input biasing so we used a large resistor value in the feedback just to bias the input voltage

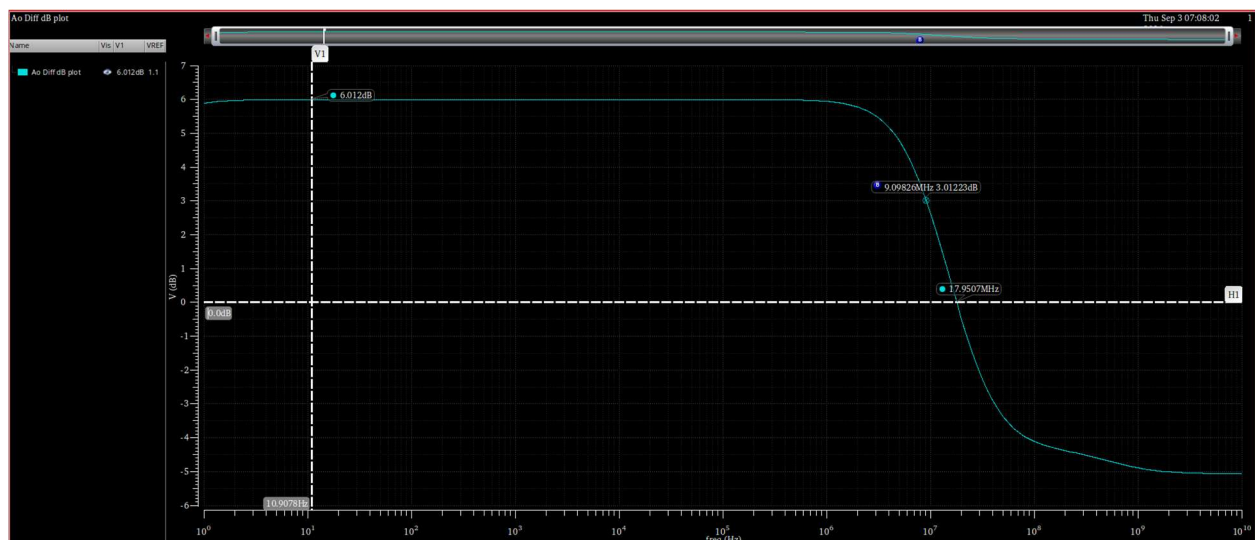
Differential closed-loop response

VODIFF vs frequency

Magnitude

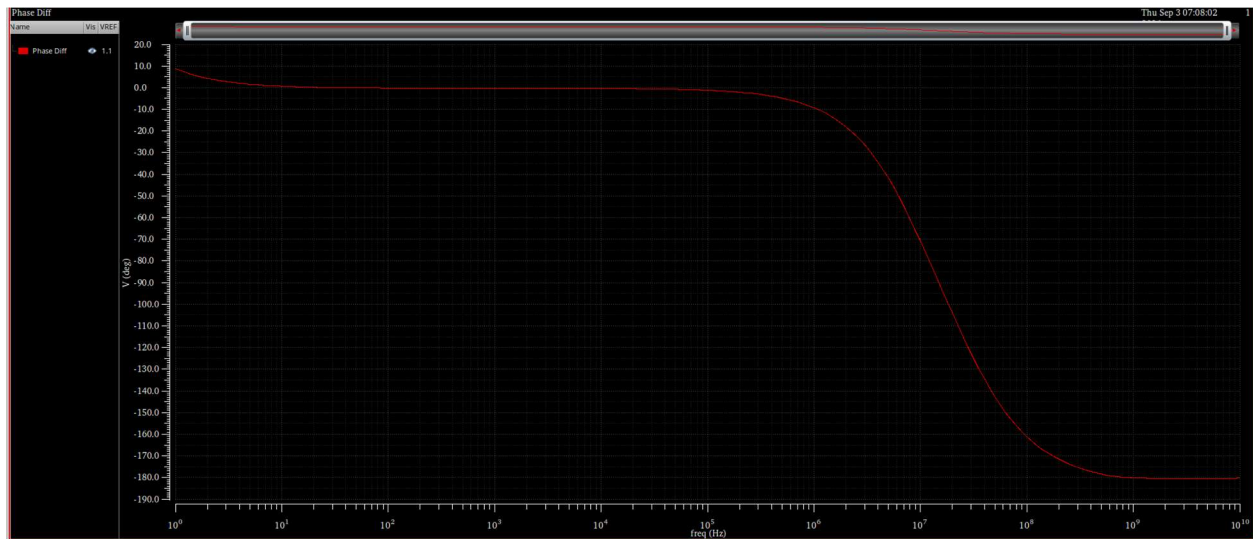


In dB





Phase



Circuit parameters

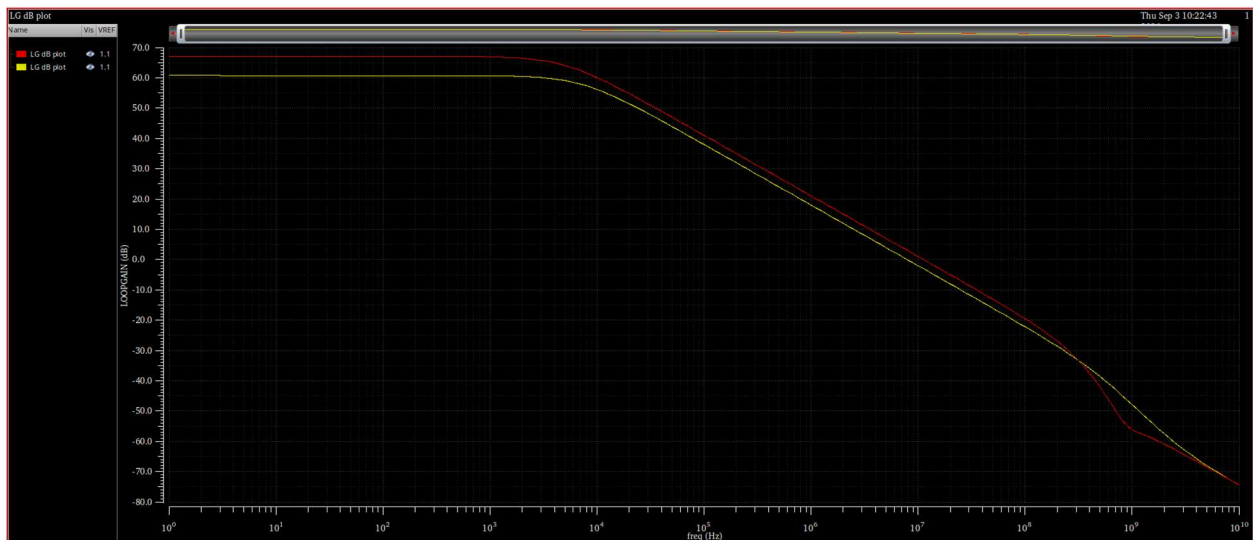
lab11:TB_CL:1	Ao Diff	1.998
lab11:TB_CL:1	Ao Diff dB	6.013
lab11:TB_CL:1	BW Diff	9.378M
lab11:TB_CL:1	UGF Diff	17.96M
lab11:TB_CL:1	GBW Diff	18.74M
lab11:TB_CL:1	PM Diff	81.21



Differential and CMFB loops stability (STB analysis)

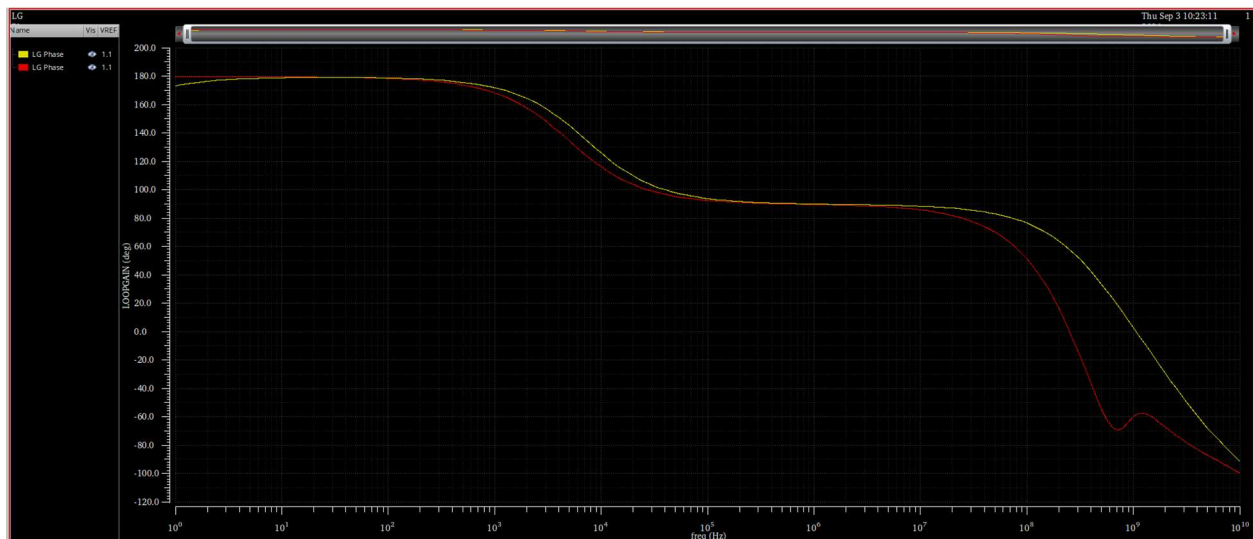
Overlaid LG

Red CM , Yellow DIFF



Overlaid Phase

Red CM , Yellow DIFF





DIFF circuit parameters

lab11:TB_CL:1	LG PM	-91.03
lab11:TB_CL:1	LG	1.126k
lab11:TB_CL:1	LG dB	61.03
lab11:TB_CL:1	LG BW	7.084k
lab11:TB_CL:1	LG UGF	8.104M
lab11:TB_CL:1	LG GBW	7.977M

CM circuit parameters

lab11:TB_CL:1	LG PM	-94.46
lab11:TB_CL:1	LG	2.32k
lab11:TB_CL:1	LG dB	67.31
lab11:TB_CL:1	LG BW	4.939k
lab11:TB_CL:1	LG UGF	11.5M
lab11:TB_CL:1	LG GBW	11.46M

GBW and PM comparison

	CM	DIFF
GBW	11.46M	7.977M
PM	85.54	88.97

Comment

the difference is due to that the CM gain is higher which increases the GBW and the PM is also better because so the second pole is at a higher frequency as there is no additional HIN due the CMFB circuit

LG and GBW comparison between OL and CL

	OL	CL
GBW	56.69M	7.977M
LG	3.352k	1.126k



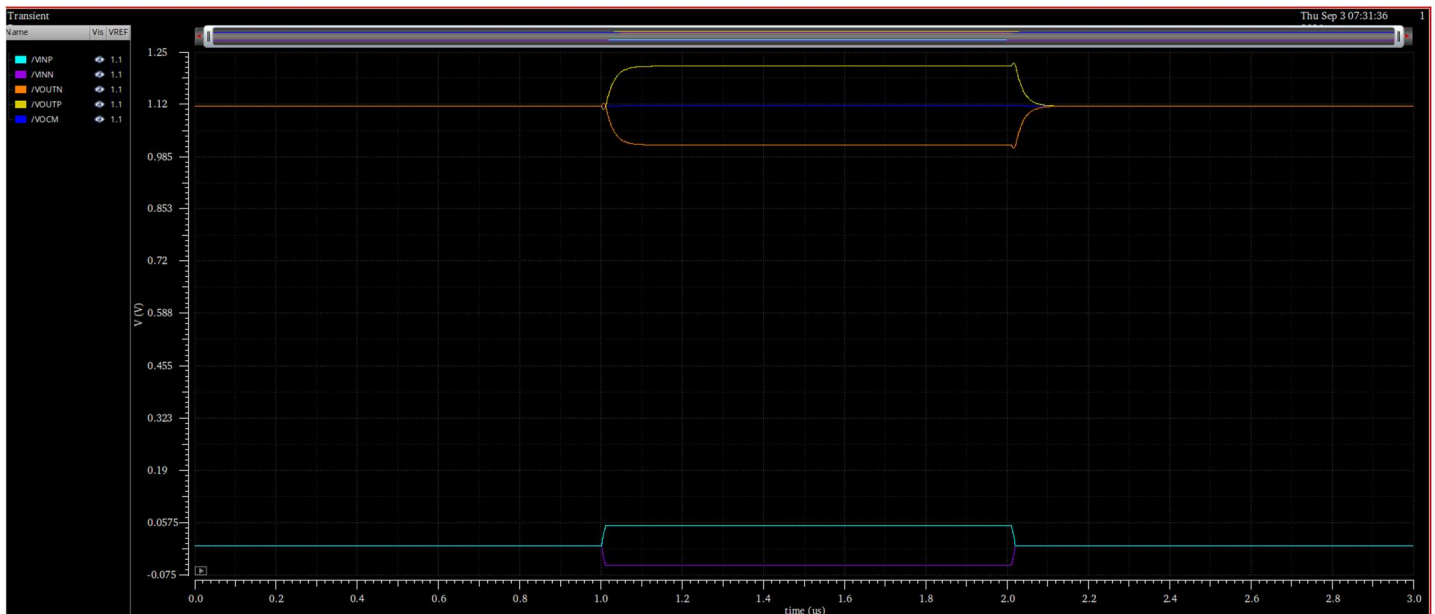
Comment

The lower gain is due to the loading effect from the input and the feedback caps as the Aol loaded is $\frac{2}{3}$ the unloaded open loop and the beta in case of considering the returned voltage will be $\frac{1}{3}$ that's why the LG is third the Aol unloaded and the GBW is also lower due to the lower gain and to the capacitive load at the output as the CF and CIN load the output

PART 6: Closed Loop Simulation (Transient Analysis)

Differential and CMFB loops stability (transient analysis) (Differential impulse)

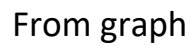
voutN ,voutp ,vinp, vinn&vocm



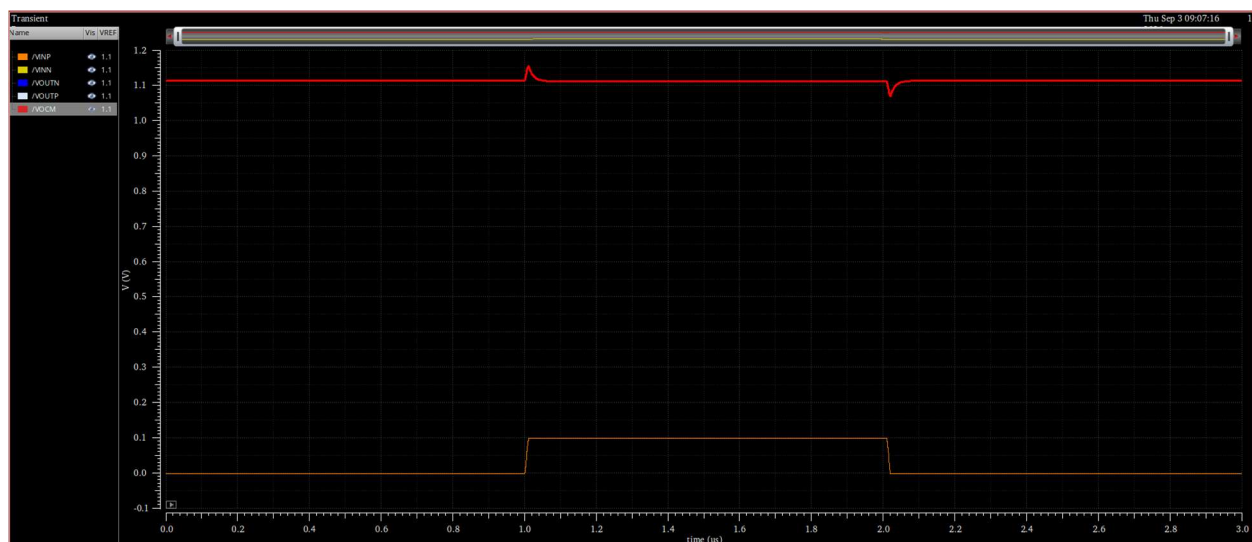
There is no ringing in both the CM and DIFF as both have an adequate above 80 PM

Tsettle

lab11:TB_CL:1	T Rise	88.56n
lab11:TB_CL:1	settling time	99.02n



Differential and CMFB loops stability (transient analysis): CM input pulse

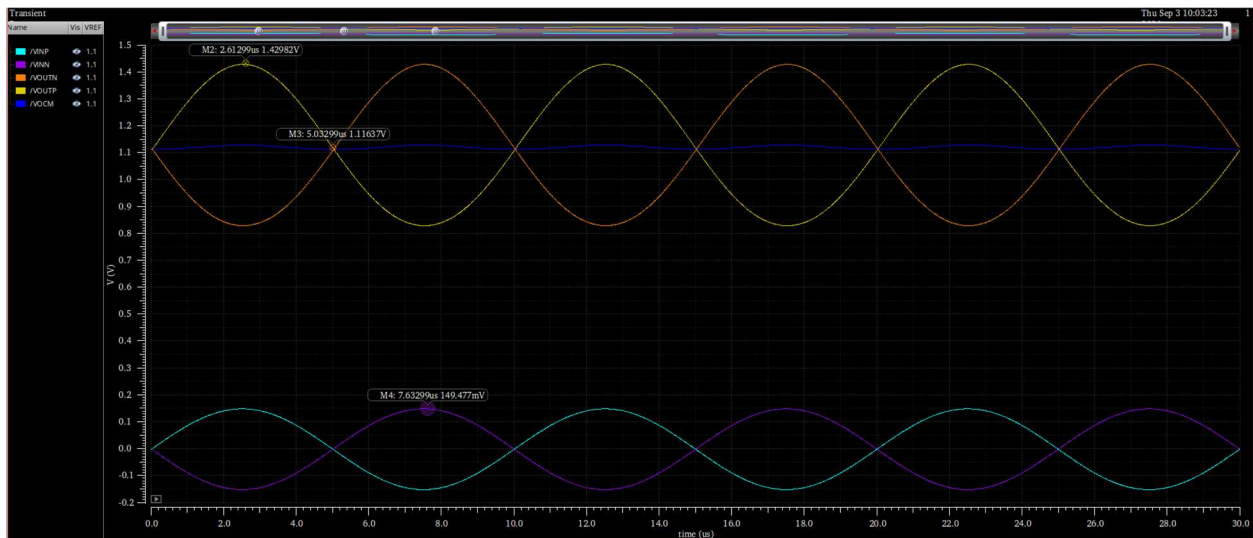




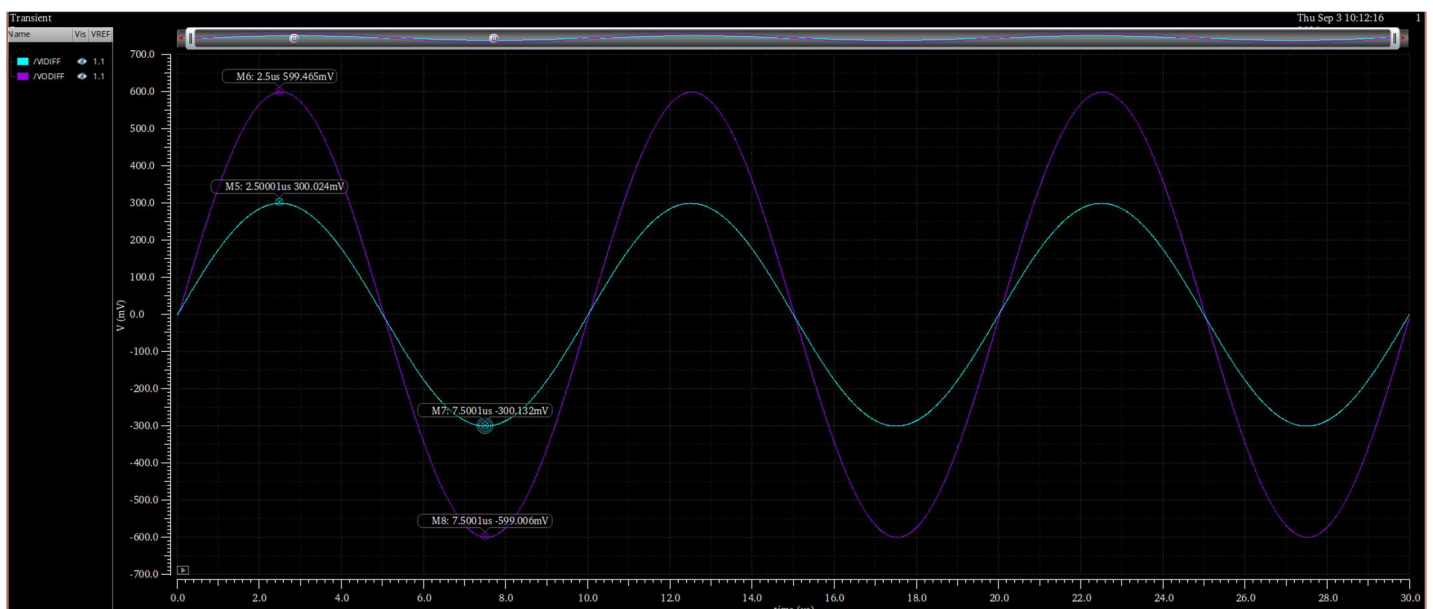
There is no ringing just a spike until the CMFB circuit adjusted the V_{OCM} level again as the PM for both Diff and CM loop above 80

Output swing

V_{INP}, V_{INN}, V_{OUTP}, V_{OUTN}, and V_{OCM} overlaid



V_{IDIFF} and V_{ODIFF}





Calculations

$$V_{ID,pp} = V_{ID,max} - V_{ID,min}$$

$$V_{ID,pp} = 300.024 \text{ mV} - (-300.132 \text{ mV}) = 600.156 \text{ mV}$$

$$V_{OD,pp} = V_{OD,max} - V_{OD,min}$$

$$V_{OD,pp} = 599.465 \text{ mV} - (-599.006 \text{ mV}) = 1.198471 \text{ V}$$

$$|A_{CL}| = \frac{V_{OD,pp}}{V_{ID,pp}} = \frac{1.198471}{0.600156} = 1.9969 \text{ V/V} \approx 2 \text{ V/V}$$